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FURTHER MATHEMATICS • UPPER SIXTH

UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 18

SPECIAL DISCRETE PROBABILITY DISTRIBUTIONS

SUGGESTED TIME: 11 PERIODS

IN THIS CHAPTER

This chapter studies the four fundamental discrete probability distributions of the GCE Further Mathematics syllabus: binomial, uniform, geometric and Poisson. For each one, we derive the mean and variance, learn to use cumulative tables, and apply the models to real-life counting problems, including the Poisson approximation of the binomial.

- By the end of this chapter I can recognise a binomial situation and state its parameters n and p.
- By the end of this chapter I can compute binomial probabilities and derive/use E(X)=np and Var(X)=npq.
- By the end of this chapter I can read cumulative binomial probability tables to find P(X≤r), P(X≥r) and P(a≤X≤b).
- By the end of this chapter I can use the recurrence formula P(X=r+1)/P(X=r) to locate the mode of a binomial distribution.
- By the end of this chapter I can define the discrete uniform distribution and derive its expectation (n+1)/2 and variance (n²−1)/12.
- By the end of this chapter I can model 'waiting time to first success' with the geometric distribution and use E(X)=1/p and Var(X)=q/p².

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SUMMARY SHEET the essentials to remember

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CHAPTER OBJECTIVES

- By the end of this chapter I can recognise a binomial situation and state its parameters n and p.
- By the end of this chapter I can compute binomial probabilities and derive/use $E(X)=np$ and $Var(X)=npq$.
- By the end of this chapter I can read cumulative binomial probability tables to find $P(X\le r)$, $P(X\ge r)$ and $P(a\le X\le b)$.
- By the end of this chapter I can use the recurrence formula $P(X=r+1)/P(X=r)$ to locate the mode of a binomial distribution.
- By the end of this chapter I can define the discrete uniform distribution and derive its expectation $(n+1)/2$ and variance $(n^2-1)/12$.
- By the end of this chapter I can model 'waiting time to first success' with the geometric distribution and use $E(X)=1/p$ and $Var(X)=q/p^2$.
- By the end of this chapter I can model rare random events with the Poisson distribution, use $E(X)=Var(X)=\lambda$, and adjust λ to the unit interval.
- By the end of this chapter I can approximate a binomial distribution by a Poisson distribution when n is large and p is small (np moderate).
- By the end of this chapter I can use cumulative Poisson tables and the additive property of independent Poisson variables.
- By the end of this chapter I can relate a probability distribution to an observed frequency distribution and compare theoretical and experimental means.

PREREQUISITES

- Probability of events, independence and the binomial coefficients nCr (Lower Sixth Pure).
- Random variables, probability distributions, expectation E(X) and variance Var(X) (Lower Sixth Statistics).
- Summation of series: sum of first n integers, sum of squares, geometric series (Lower Sixth Pure).
- The exponential function e^x and its use in limits (Lower Sixth Pure).
- Factorials and manipulation of algebraic fractions.

1 THE BINOMIAL DISTRIBUTION: PROBABILITIES, MEAN AND VARIANCE

At the Mfoundi market in Yaoundé, a trader who sells 200 avocados a day knows from experience that about 3 out of every 10 are overripe and unsellable. How many bad avocados should she expect each morning, and how likely is she to lose more than 70 in one day? Meanwhile, the MTN call centre in Douala receives on average 4 calls per minute: what is the probability of receiving no call at all in the next minute? And a voter queue at a polling station in Bamenda raises another question: how many people must be checked before the first registered voter is found? These everyday situations are modelled by the **special discrete distributions** of this chapter: **binomial**, **uniform**, **geometric** and **Poisson**. In this first section we build the most fundamental of them all, the binomial distribution, from the ground up: we define it, compute its probabilities, and derive its mean and variance rigorously.

1.1 Bernoulli Trials: the Building Blocks

Imagine tossing a coin once and recording only whether it lands Heads. Or inspecting one avocado and recording only whether it is overripe. Each of these experiments has exactly two possible outcomes, which we conventionally call **success** and **failure**. Such an experiment is called a **Bernoulli trial**.

DEFINITION **Bernoulli trial**

A Bernoulli trial is a random experiment with exactly two outcomes: success (S), with probability p, and failure (F), with probability $q = 1 - p$. A sequence of Bernoulli trials is a sequence of independent repetitions of this experiment, with p staying constant from trial to trial.

The word "success" is a label, not a value judgement: if we are counting overripe avocados, then "success" means "the avocado is overripe". What matters mathematically is the structure, not the story.

METHOD **Recognising a binomial situation**

To decide whether a random variable X follows a binomial distribution, check the four conditions:

1. Fixed number of trials: the experiment is repeated a fixed number n of times.
2. Two outcomes: each trial results in success or failure only.
3. Constant probability: the probability of success p is the same for every trial.
4. Independence: the outcome of one trial does not affect the others.

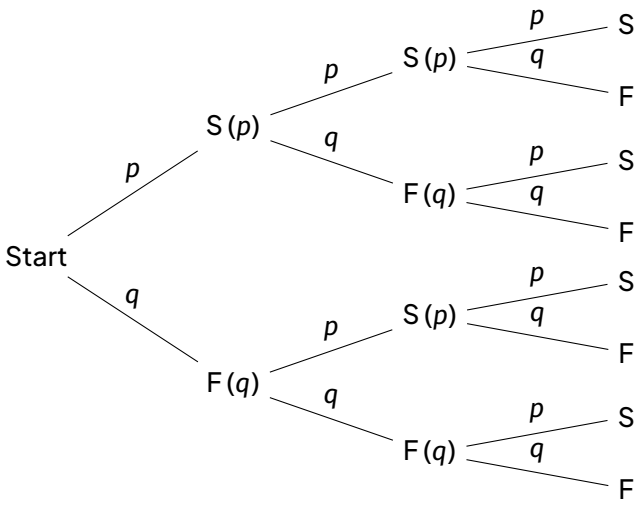
If all four hold and X counts the number of successes, then X is binomial.

WARNING **Sampling without replacement**

Drawing items from a small finite population without replacement breaks the independence and constant-p conditions. For example, picking 5 students from a class of 30 and counting the number of girls is not exactly binomial (it is hypergeometric). However, when the population is very large compared with the sample (e.g. 5 bolts from a production line of thousands), the binomial model is an excellent approximation and examiners accept it.

1.2 Definition and Probability Formula of $X \sim B(n, p)$

Suppose we perform $n = 3$ independent Bernoulli trials with $P(S) = p$, and we want the probability of exactly 2 successes. The tree diagram below shows all $2^3 = 8$ possible outcomes.



Tree diagram of 3 Bernoulli trials. Exactly 2 successes occur along the paths SSF, SFS and FSS: there are $\binom{3}{2} = 3$ such paths, each of probability p^2q.

Each path containing exactly 2 successes and 1 failure has probability p^2q (multiplication rule, by independence), and the number of such paths is the number of ways of choosing which 2 of the 3 trials are successes, namely $\binom{3}{2} = 3$. Hence

$$P(\text{exactly 2 successes}) = \binom{3}{2} p^2 q.$$

This reasoning generalises immediately.

DEFINITION **Binomial distribution**

A discrete random variable X follows a binomial distribution with parameters n and p, written $X \sim B(n, p)$, if X counts the number of successes in n independent Bernoulli trials with $P(\text{success}) = p$. Its probability mass function is

$$P(X = r) = \binom{n}{r} p^r q^{n-r}, \qquad r = 0, 1, 2, \ldots, n, \quad q = 1 - p.$$

The name "binomial" comes from the binomial theorem: summing all the probabilities gives

$$\sum_{r=0}^{n} \binom{n}{r} p^r q^{n-r} = (p + q)^n = 1^n = 1,$$

which confirms that the formula defines a genuine probability distribution.

WARNING **Do not forget the coefficient**

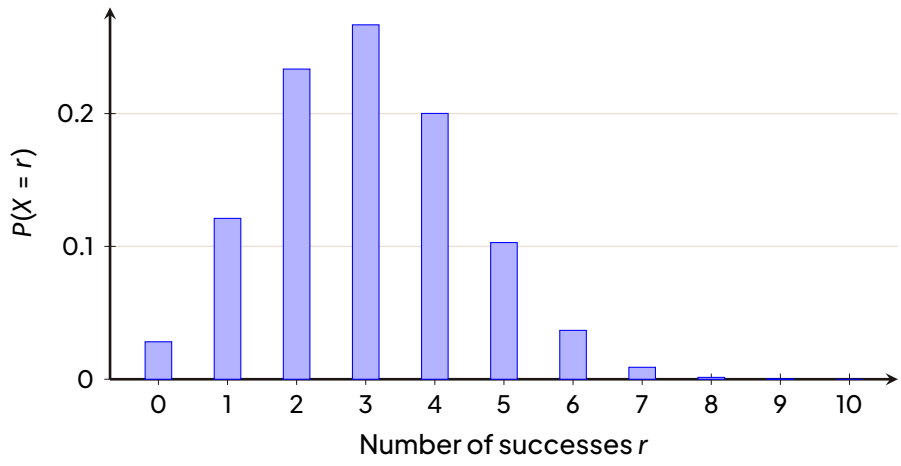
A very common mistake is to write $P(X = r) = p^r q^{n-r}$, forgetting $\binom{n}{r}$. The factor $p^r q^{n-r}$ is the probability of one particular arrangement of r successes and $n - r$ failures; you must multiply by the number of arrangements, $\binom{n}{r}$.

EXEMPLE **Building the full distribution table**

A fair coin is tossed 4 times and X counts the number of Heads. Then $X \sim B(4, 0.5)$ and $P(X = r) = \binom{4}{r}(0.5)^4 = \binom{4}{r}/16$:

r	0	1	2	3	4
$P(X = r)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Check: $\frac{1+4+6+4+1}{16} = 1$. ✓



Probability mass function of $X \sim B(10, 0.3)$. The distribution is skewed to the right because $p < 0.5$; the most likely value is $r = 3 = np$.

The bar chart of $B(10, 0.3)$ shows a typical shape: the probabilities rise to a peak near np and then fall away. When $p = 0.5$ the distribution is symmetric; when $p < 0.5$ it has a tail to the right; when $p > 0.5$ a tail to the left.

1.3 Derivation of the Mean: $E(X) = np$

Intuition first: if 3 avocados out of 10 are bad, then out of 200 avocados we "expect" $200 \times 0.3 = 60$ bad ones. Expectation should be n times p. Let us prove it.

PROPERTY **Mean of the binomial distribution — proof examinable**

If $X \sim B(n, p)$, then

$$E(X) = np.$$

Proof. By definition, $E(X) = \sum_{r=0}^{n} r\,P(X = r) = \sum_{r=0}^{n} r\binom{n}{r}p^r q^{n-r}$. The term $r = 0$ vanishes. For $r \geq 1$ we use the key identity

$$r\binom{n}{r} = r \cdot \frac{n!}{r!(n-r)!} = \frac{n!}{(r-1)!(n-r)!} = n \cdot \frac{(n-1)!}{(r-1)!\,(n-r)!} = n\binom{n-1}{r-1}.$$

Therefore

$$E(X) = \sum_{r=1}^{n} n\binom{n-1}{r-1}p^r q^{n-r} = np \sum_{r=1}^{n} \binom{n-1}{r-1}p^{r-1}q^{(n-1)-(r-1)}.$$

Putting $s = r - 1$, the sum becomes $\sum_{s=0}^{n-1}\binom{n-1}{s}p^s q^{(n-1)-s} = (p+q)^{n-1} = 1$. Hence $E(X) = np$. ■

1.4 Derivation of the Variance: $\mathrm{Var}(X) = npq$

For the variance we use $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$, but $E(X^2)$ is awkward to compute directly. The trick is to compute the factorial moment $E\big(X(X-1)\big)$ first, because $r(r-1)\binom{n}{r} = n(n-1)\binom{n-2}{r-2}$.

PROPERTY **Variance of the binomial distribution — proof examinable**

If $X \sim B(n, p)$, then

$$\mathrm{Var}(X) = npq, \qquad \text{and} \qquad \sigma = \sqrt{npq}.$$

Proof. First,

$$E\big(X(X-1)\big) = \sum_{r=0}^{n} r(r-1)\binom{n}{r}p^r q^{n-r}.$$

The terms $r = 0$ and $r = 1$ vanish. For $r \geq 2$,

$$r(r-1)\binom{n}{r} = \frac{n!}{(r-2)!(n-r)!} = n(n-1)\binom{n-2}{r-2}.$$

Hence, factoring out $n(n-1)p^2$ and setting $s = r - 2$,

$$E\big(X(X-1)\big) = n(n-1)p^2 \sum_{r=2}^{n}\binom{n-2}{r-2}p^{r-2}q^{n-r}$$

$$= n(n-1)p^2 \sum_{s=0}^{n-2}\binom{n-2}{s}p^s q^{(n-2)-s} = n(n-1)p^2(p+q)^{n-2} = n(n-1)p^2.$$

Now $E(X^2) = E\big(X(X-1)\big) + E(X) = n(n-1)p^2 + np$, so

$$\mathrm{Var}(X) = n(n-1)p^2 + np - (np)^2 = n^2p^2 - np^2 + np - n^2p^2$$
$$= np(1-p) = npq. \qquad \blacksquare$$

KEY POINTS **Key results**

If $X \sim B(n, p)$ with $q = 1 - p$:

$$P(X = r) = \binom{n}{r}p^r q^{n-r}, \qquad E(X) = np, \qquad \mathrm{Var}(X) = npq.$$

The mean tells you the expected number of successes; the variance measures the spread around that expectation. The variance is greatest when $p = 0.5$ (maximum uncertainty) and tends to 0 as $p \to 0$ or $p \to 1$.

WORKED EXAMPLE 1 **The avocado trader**

The Mfoundi trader sells $n = 200$ avocados daily, each independently overripe with probability $p = 0.3$. Let X be the number of unsellable avocados.

1. State the distribution of X.
2. Find the expected number of bad avocados and the standard deviation.
3. Compute $P(X = 60)$.

SOLUTION

Solution.

1. Fixed $n = 200$ trials, two outcomes (overripe or not), constant $p = 0.3$, independence assumed: $X \sim B(200, 0.3)$.
2. $E(X) = np = 200 \times 0.3 = 60$ bad avocados per day. $\text{Var}(X) = npq = 200 \times 0.3 \times 0.7 = 42$, so $\sigma = \sqrt{42} \approx 6.48$. She should expect about 60 bad avocados, typically fluctuating by roughly 6 or 7 from day to day.
3. $P(X = 60) = \binom{200}{60}(0.3)^{60}(0.7)^{140} \approx 0.0615$. Even the most likely single value has only about a 6% chance — with 201 possible values, the probability is spread thinly.

1.5 Finding n and p from the Mean and Variance

In many GCE questions, n and p are unknown but $E(X)$ and $\text{Var}(X)$ are given. This yields two simultaneous equations:

$$np = \mu, \qquad npq = \sigma^2.$$

METHOD **Recovering n and p**

1. Divide the variance by the mean: $\dfrac{npq}{np} = q = \dfrac{\sigma^2}{\mu}$.
2. Deduce $p = 1 - q$.
3. Deduce $n = \dfrac{\mu}{p}$ (check that n is a positive integer).

WORKED EXAMPLE 2 **Unknown n and p**

A binomial random variable X has mean 6 and variance 4.2. Find n, p and $P(X = 2)$.

SOLUTION

Solution. We have $np = 6$ and $npq = 4.2$.

$$q = \frac{npq}{np} = \frac{4.2}{6} = 0.7 \quad \Longrightarrow \quad p = 1 - 0.7 = 0.3, \qquad n = \frac{6}{0.3} = 20.$$

So $X \sim B(20, 0.3)$ and

$$P(X = 2) = \binom{20}{2}(0.3)^2(0.7)^{18} = 190 \times 0.09 \times (0.7)^{18} \approx 0.0278.$$

WARNING **Sanity checks**

Always verify that your answers make sense: $0 < p < 1$, n a positive integer, and $\mathrm{Var}(X) < E(X)$ for a binomial (since $q < 1$). If a question gives a variance larger than the mean, the variable cannot be binomial — this observation will matter when we meet the Poisson distribution, where mean and variance are equal.

EXERCISE 1 **Exercises on the Binomial Distribution**

1. A fair die is rolled 6 times. Let X be the number of sixes obtained. State the distribution of X and compute $P(X = 2)$, $E(X)$ and $\mathrm{Var}(X)$.
2. It is known that 70% of candidates in a certain school pass the GCE Advanced Level. A random sample of 8 candidates is taken. Find the probability that exactly 5 pass, and the expected number of passes.
3. A binomial variable has mean 12 and variance 3. Find n, p and $P(X = 12)$.

SOLUTION **Exercise 1**

1. $X \sim B(6, \frac{1}{6})$; $P(X = 2) = \binom{6}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^4 = 15 \times \frac{1}{36} \times \frac{625}{1296} \approx 0.2009$; $E(X) = 6 \times \frac{1}{6} = 1$; $\mathrm{Var}(X) = 6 \times \frac{1}{6} \times \frac{5}{6} = \frac{5}{6} \approx 0.833$.
2. $X \sim B(8, 0.7)$; $P(X = 5) = \binom{8}{5}(0.7)^5(0.3)^3 = 56 \times 0.16807 \times 0.027 \approx 0.2541$; $E(X) = 5.6$.
3. $q = \frac{3}{12} = 0.25$, so $p = 0.75$ and $n = \frac{12}{0.75} = 16$. $P(X = 12) = \binom{16}{12}(0.75)^{12}(0.25)^4 = 1820 \times (0.75)^{12} \times \frac{1}{256} \approx 0.2252$.

DID YOU KNOW? **Jacob Bernoulli**

The Bernoulli trial is named after the Swiss mathematician Jacob Bernoulli (1655–1705), whose book Ars Conjectandi laid the foundations of probability theory. The same family produced no fewer than eight distinguished mathematicians. The binomial distribution you have just studied is the engine behind opinion polls, quality control in factories, and medical trials — including the analysis of examination pass rates in Cameroonian schools.

2 CUMULATIVE BINOMIAL TABLES AND THE MODE

In the previous section we derived the probability mass function of the binomial distribution $B(n, p)$ and computed its mean $\mu = np$ and variance $\sigma^2 = npq$. While the formula $P(X = r) = \binom{n}{r} p^r q^{n-r}$ is exact, evaluating it by hand for several values of r becomes tedious, especially when n is large. In examinations and in practice, statisticians use **cumulative binomial tables** which give $P(X \le r)$ directly. This section teaches you how to read those tables efficiently, how to handle probabilities that are not directly tabulated, and how to locate the mode of the distribution using a simple recurrence relation.

2.1 Structure of Cumulative Binomial Tables

Standard GCE cumulative binomial tables are organised as follows:

- Rows correspond to the number of trials n (typically $n = 1, 2, \ldots, 50$).
- Columns correspond to the success probability p (usually $p = 0.05, 0.10, \ldots, 0.50$).
- Inside the table, for a given n and p, you find a list of values $P(X \le r)$ for $r = 0, 1, \ldots, n$.

Only probabilities for $p \le 0.5$ are printed. If $p > 0.5$ we use the symmetry of the binomial distribution (see below).

n		$p = 0.05$	$p = 0.10$	$p = 0.20$	$p = 0.30$	$p = 0.40$	$p = 0.50$
8	0	0.6634	0.4305	0.1678	0.0576	0.0168	0.0039
8	1	0.9428	0.8131	0.5033	0.2553	0.1064	0.0352
8	2	0.9942	0.9619	0.7969	0.5518	0.3154	0.1445
8	3	0.9994	0.9950	0.9437	0.8059	0.5941	0.3633
8	4	0.9999	0.9995	0.9896	0.9420	0.8263	0.6367
8	5	1.0000	0.9999	0.9988	0.9887	0.9502	0.8555
8	6	1.0000	1.0000	0.9999	0.9988	0.9930	0.9648
8	7	1.0000	1.0000	1.0000	0.9999	0.9997	0.9961
8	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

$\rightarrow P(X \le 3) = 0.8059$

Extract of a cumulative binomial table for $n = 8$. The highlighted cell gives $P(X \le 3)$ when $p = 0.30$.

2.2 Reading Probabilities from the Table

The table gives **cumulative probabilities** $P(X \le r)$. From these we can obtain any probability we need.

METHOD **Using Cumulative Binomial Tables**

Step 1: Identify n and p. If $p > 0.5$, see the symmetry trick below.
Step 2: Locate the block for your n, then the column for your p.
Step 3: Read $P(X \le r)$ directly from the row corresponding to r.
Step 4: Convert your required probability to the form $P(X \le r)$ using the rules:

- $P(X = r) = P(X \le r) - P(X \le r - 1)$ (with $P(X \le -1) = 0$).
- $P(X \ge r) = 1 - P(X \le r - 1)$.
- $P(X > r) = 1 - P(X \le r)$.
- $P(a \le X \le b) = P(X \le b) - P(X \le a - 1)$.

WARNING **Strict vs Non-Strict Inequalities — The #1 GCE Trap**

Many candidates write $P(X < r)$ as $P(X \le r)$ instead of $P(X \le r - 1)$. Remember:

$$P(X < r) = P(X \le r - 1), \qquad P(X > r) = 1 - P(X \le r).$$

Always convert strict inequalities to non-strict ones before looking at the table.

2.3 Worked Examples: Reading the Table

WORKED EXAMPLE 3 **Using the table for** $X \sim B(8, 0.3)$

Using the table extract above, find:

1. $P(X = 3)$
2. $P(X \ge 4)$
3. $P(2 \le X \le 5)$

SOLUTION

Solution. From the table, for $n = 8$, $p = 0.30$:

r	$P(X \le r)$
0	0.0576
1	0.2553
2	0.5518
3	0.8059
4	0.9420
5	0.9887

1. $P(X = 3) = P(X \le 3) - P(X \le 2) = 0.8059 - 0.5518 = 0.2541$.
2. $P(X \ge 4) = 1 - P(X \le 3) = 1 - 0.8059 = 0.1941$.
3. $P(2 \le X \le 5) = P(X \le 5) - P(X \le 1) = 0.9887 - 0.2553 = 0.7334$.

2.4 The Symmetry Trick for $p > 0.5$

Tables only go up to $p = 0.5$. If $X \sim B(n, p)$ with $p > 0.5$, define $Y = n - X$, the number of failures. Then $Y \sim B(n, q)$ where $q = 1 - p < 0.5$. Any probability about X can be rewritten in terms of Y and read from the table.

PROPERTY **Symmetry of the Binomial Distribution**

If $X \sim B(n, p)$ and $Y = n - X$, then $Y \sim B(n, 1 - p)$. Consequently:

$$P(X \le r) = P(Y \ge n - r) = 1 - P(Y \le n - r - 1).$$

EXEMPLE **Example:** $p > 0.5$

Let $X \sim B(10, 0.7)$. Find $P(X \le 8)$ using tables.

Here $p = 0.7 > 0.5$. Let $Y = 10 - X \sim B(10, 0.3)$.

$$P(X \le 8) = P(10 - Y \le 8) = P(Y \ge 2) = 1 - P(Y \le 1).$$

Look up $n = 10$, $p = 0.30$, $r = 1$ in the table to get $P(Y \le 1)$, then subtract from 1.

2.5 The Recurrence Formula and the Mode

Instead of computing each $P(X = r)$ from the factorial formula, we can generate the whole distribution recursively.

PROPERTY **Recurrence Relation for Binomial Probabilities**

For $X \sim B(n, p)$ with $q = 1 - p$,

$$\frac{P(X = r+1)}{P(X = r)} = \frac{(n-r)p}{(r+1)q}, \qquad r = 0, 1, \ldots, n-1.$$

Proof:

$$\frac{P(X = r+1)}{P(X = r)} = \frac{\binom{n}{r+1} p^{r+1} q^{n-r-1}}{\binom{n}{r} p^r q^{n-r}} = \frac{\frac{n!}{(r+1)!(n-r-1)!}}{\frac{n!}{r!(n-r)!}} \cdot \frac{p}{q} = \frac{r!(n-r)!}{(r+1)!(n-r-1)!} \cdot \frac{p}{q} = \frac{n-r}{r+1} \cdot \frac{p}{q}.$$

KEY POINTS **Generating the Distribution Recursively**

Start with $P(X = 0) = q^n$. Then for $r = 0, 1, \ldots, n-1$:

$$P(X = r+1) = P(X = r) \times \frac{(n-r)p}{(r+1)q}.$$

This avoids large factorials and is ideal for programming or manual calculation when tables are unavailable.

2.6 Finding the Mode from the Recurrence Ratio

The **mode** is the value of r for which $P(X = r)$ is maximal. The ratio $\frac{P(X=r+1)}{P(X=r)}$ tells us whether the probabilities are increasing or decreasing.

- If the ratio > 1, then $P(X = r+1) > P(X = r)$ — probabilities are increasing.
- If the ratio < 1, then $P(X = r+1) < P(X = r)$ — probabilities are decreasing.
- If the ratio $= 1$, then $P(X = r+1) = P(X = r)$ — two adjacent values are equally maximal.

The ratio exceeds 1 when

$$\frac{(n-r)p}{(r+1)q} > 1 \iff (n-r)p > (r+1)q \iff r < (n+1)p - 1.$$

Thus probabilities increase while $r < (n+1)p - 1$ and decrease after $r > (n+1)p - 1$. The peak occurs at the integer part of $(n+1)p$.

PROPERTY **Mode of the Binomial Distribution**

Let $X \sim B(n, p)$ and let $m = (n+1)p$.

- If m is not an integer, the unique mode is $\lfloor m \rfloor$ (the integer part of m).

- If m is an integer, there are two modes: m and $m-1$ (both have the same maximal probability).

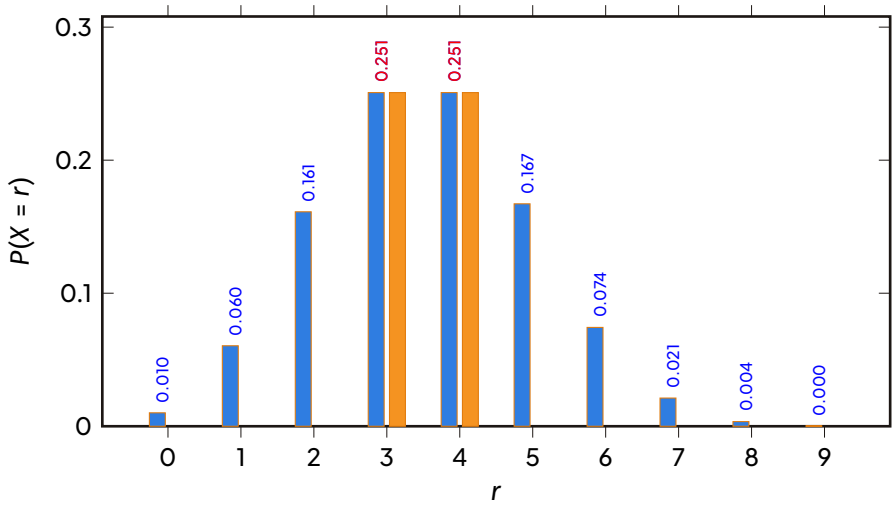
EXEMPLE **Mode of $B(9, 0.4)$**

Here $n = 9$, $p = 0.4$, so $m = (9 + 1) \times 0.4 = 4$. Since $m = 4$ is an integer, the distribution has two modes: $r = 4$ and $r = 3$.

We can verify using the recurrence ratio:

$$\frac{P(X = 4)}{P(X = 3)} = \frac{(9 - 3) \times 0.4}{4 \times 0.6} = \frac{2.4}{2.4} = 1.$$

Hence $P(X = 4) = P(X = 3)$ and both are maximal.



Probability mass function of $B(9, 0.4)$. The two modal bars at $r = 3$ and $r = 4$ are highlighted in orange.

2.7 Worked Example: Full Distribution and Mode

WORKED EXAMPLE 4 **Constructing $B(6, 0.3)$ recursively and finding its mode**

1. Generate all probabilities $P(X = r)$ for $r = 0, \ldots, 6$ using the recurrence formula.
2. State the mode.

SOLUTION

Solution.

1. $q = 0.7$. $P(X = 0) = 0.7^6 = 0.117649$.

$$P(X = 1) = 0.117649 \times \frac{6 \times 0.3}{1 \times 0.7} = 0.302526$$
$$P(X = 2) = 0.302526 \times \frac{5 \times 0.3}{2 \times 0.7} = 0.324135$$
$$P(X = 3) = 0.324135 \times \frac{4 \times 0.3}{3 \times 0.7} = 0.185220$$
$$P(X = 4) = 0.185220 \times \frac{3 \times 0.3}{4 \times 0.7} = 0.059535$$
$$P(X = 5) = 0.059535 \times \frac{2 \times 0.3}{5 \times 0.7} = 0.010206$$
$$P(X = 6) = 0.010206 \times \frac{1 \times 0.3}{6 \times 0.7} = 0.000729$$

(Sum ≈ 1.0000, a good check.)

2. $m = (6+1) \times 0.3 = 2.1$. Not an integer ⇒ unique mode at $\lfloor 2.1 \rfloor = 2$. Indeed $P(X = 2) = 0.3241$ is the largest probability.

2.8 Linking Probability Distributions to Frequency Distributions

In Lesson 133 you will see that a theoretical probability distribution (like $B(n, p)$) can be fitted to an observed frequency distribution. The expected frequency for value r is $N \times P(X = r)$ where N is the total number of observations. The recurrence formula is extremely useful here: once you have $P(X = 0)$, you can compute all expected frequencies recursively without recalculating binomial coefficients.

EXEMPLE **Fitting a Binomial Distribution**

A die is rolled 120 times and the number of sixes in each set of 5 rolls is recorded. If the die is fair, the number of sixes per set follows $B(5, 1/6)$. Use the recurrence formula to find the expected frequencies for 0,1,2,3,4,5 sixes.

$p = 1/6$, $q = 5/6$, $N = 120$. $P(X = 0) = (5/6)^5 \approx 0.4019$; expected frequency $= 120 \times 0.4019 \approx 48.2$. Recurrence ratio $= \frac{(5-r)\times 1/6}{(r+1)\times 5/6} = \frac{5-r}{5(r+1)}$. Compute successively: $r = 0 \to 1$: ratio $= 5/5 = 1$ ⇒ $P(X = 1) = 0.4019$ (freq 48.2) $r = 1 \to 2$: ratio $= 4/10 = 0.4$ ⇒ $P(X = 2) = 0.1608$ (freq 19.3) $r = 2 \to 3$: ratio $= 3/15 = 0.2$ ⇒ $P(X = 3) = 0.0322$ (freq 3.9) $r = 3 \to 4$: ratio $= 2/20 = 0.1$ ⇒ $P(X = 4) = 0.0032$ (freq 0.4) $r = 4 \to 5$: ratio $= 1/25 = 0.04$ ⇒ $P(X = 5) = 0.0001$ (freq 0.0)

2.9 Summary of Key Points

KEY POINTS **Cumulative Tables and Mode — Key Points**

- Tables give $P(X \le r)$ for $p \le 0.5$. Convert all probabilities to this form.
- $P(X = r) = P(X \le r) - P(X \le r-1)$; $P(X \ge r) = 1 - P(X \le r-1)$; $P(a \le X \le b) = P(X \le b) - P(X \le a-1)$.
- For $p > 0.5$, use $Y = n - X \sim B(n, 1-p)$ and read from the q-column.
- Recurrence: $\dfrac{P(X = r+1)}{P(X = r)} = \dfrac{(n-r)p}{(r+1)q}$. Start from $P(X = 0) = q^n$.
- Mode = $\lfloor (n+1)p \rfloor$; if $(n+1)p$ is an integer, modes are $(n+1)p$ and $(n+1)p - 1$.

EXERCISE 2 **Practice Exercise**

$X \sim B(12, 0.35)$.

1. Using tables, find $P(X \le 4)$, $P(X = 5)$, $P(X \ge 7)$, $P(3 \le X \le 8)$.
2. If tables only go up to $p = 0.5$, explain how you would find $P(X \le 8)$ for $Y \sim B(12, 0.65)$.
3. Find the mode of X using the formula. Verify with the recurrence ratio.
4. Generate $P(X = 0)$, $P(X = 1)$, $P(X = 2)$ recursively.

SOLUTION **Exercise 1**

1. From tables ($n = 12$, $p = 0.35$): $P(X \le 4) = 0.5833$ (approx), $P(X = 5) = P(X \le 5) - P(X \le 4) = 0.7745 - 0.5833 = 0.1912$, $P(X \ge 7) = 1 - P(X \le 6) = 1 - 0.8912 = 0.1088$, $P(3 \le X \le 8) = P(X \le 8) - P(X \le 2) = 0.9737 - 0.2528 = 0.7209$.
2. For $Y \sim B(12, 0.65)$, let $Z = 12 - Y \sim B(12, 0.35)$. $P(Y \le 8) = P(12 - Z \le 8) = P(Z \ge 4) = 1 - P(Z \le 3)$. Read $P(Z \le 3)$ from $p = 0.35$ column.
3. $m = (12 + 1) \times 0.35 = 4.55$. Not integer $\Rightarrow$ mode = $\lfloor 4.55 \rfloor = 4$. Check ratio at $r = 4$: $\frac{(12-4)\times 0.35}{5 \times 0.65} = \frac{2.8}{3.25} < 1$; at $r = 3$: $\frac{9 \times 0.35}{4 \times 0.65} = \frac{3.15}{2.6} > 1$. So peak at $r = 4$.
4. $P(X = 0) = 0.65^{12} \approx 0.00569$. $P(X = 1) = 0.00569 \times \frac{12 \times 0.35}{1 \times 0.65} \approx 0.0369$. $P(X = 2) = 0.0369 \times \frac{11 \times 0.35}{2 \times 0.65} \approx 0.1088$.

2.10 Examination Tips

DID YOU KNOW? **GCE Examiner's Advice**

- Always show the conversion step: Write "$P(X > 3) = 1 - P(X \leq 3)$" before giving the numerical answer. This earns the method mark even if you misread the table.
- For $p > 0.5$: Explicitly define $Y = n - X$ and state $Y \sim B(n, 1-p)$. Do not just say "use symmetry" — show the algebra.
- Mode questions: If asked to "find the mode", state the formula $m = (n+1)p$, compute it, and give the integer part. If m is integer, write "two modes: m and $m-1$".
- Recurrence formula: If asked to "use the recurrence formula", you must show the ratio $\frac{P(X=r+1)}{P(X=r)}$ and at least one step of the calculation.

This completes the section on cumulative binomial tables and the mode. In the next section we will study the **discrete uniform distribution**, where every outcome is equally likely.

3 THE DISCRETE UNIFORM DISTRIBUTION

Having mastered the binomial distribution — its probabilities, cumulative tables, and mode — we now turn to a much simpler but equally important model: the **discrete uniform distribution**. While the binomial distribution counts successes in a fixed number of independent trials, the uniform distribution describes situations where every outcome is equally likely. Think of rolling a fair die, drawing a raffle ticket from a well-mixed drum, or generating a random digit on a calculator. In each case, symmetry dictates that no outcome is favoured over another. This section builds the distribution from first principles, derives its mean and variance rigorously, and shows how to apply it to practical problems.

3.1 Definition and Intuition

DEFINITION **Discrete Uniform Distribution on $\{1, 2, \ldots, n\}$**

A discrete random variable X follows a discrete uniform distribution on the set $\{1, 2, \ldots, n\}$ (where $n \in \mathbb{N}$, $n \geq 1$) if its probability mass function is

$$P(X = r) = \frac{1}{n} \qquad \text{for } r = 1, 2, \ldots, n,$$

and $P(X = r) = 0$ otherwise. We write $X \sim U(1, n)$.

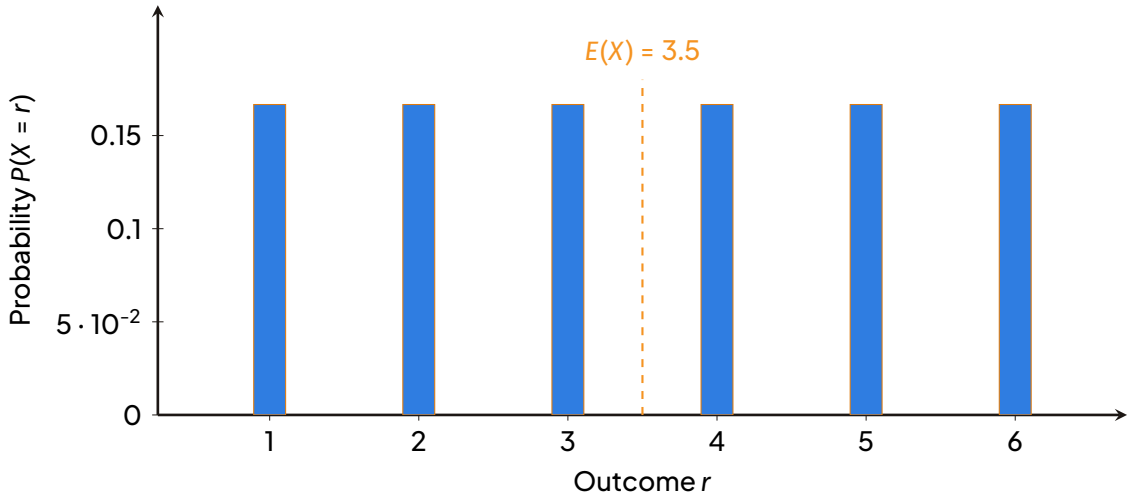
The definition captures the idea of equally likely outcomes. If an experiment has n distinct outcomes and symmetry (or careful design) guarantees that none is more probable than another,

then each outcome receives probability $1/n$. Note that the probabilities sum to 1, as required:

$$\sum_{r=1}^{n} P(X = r) = \sum_{r=1}^{n} \frac{1}{n} = n \cdot \frac{1}{n} = 1.$$

EXEMPLE **Everyday Examples**

- Fair die: A standard six-sided die has faces $\{1, 2, 3, 4, 5, 6\}$. If the die is fair, $X \sim U(1, 6)$ and $P(X = r) = 1/6$ for $r = 1, \ldots, 6$.
- Raffle ticket: A raffle at a school fate in Bamenda sells tickets numbered 1 to 1000. One ticket is drawn at random. The winning number $X \sim U(1, 1000)$.
- Random digit: A calculator's random-integer function returns an integer from 0 to 9 with equal probability. Then X is uniform on $\{0, 1, \ldots, 9\}$ — a shift of the standard definition (see the extension below).



Probability mass function of $X \sim U(1, 6)$ (fair die). All bars have equal height $1/6$; the mean $E(X) = 3.5$ is marked by the orange dashed line.

3.2 Expectation and Variance

We derive the mean and variance using two classic summation formulas. These derivations are **examinable** — you must be able to reproduce them step by step in the GCE Advanced Level examination.

METHOD **Key Summation Formulas to Memorise**

For any positive integer n:

1. $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$ (sum of the first n integers)

2. $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$ (sum of the first n squares)

These two formulas drive the derivations of $E(X)$ and $E(X^2)$ for the uniform distribution.

PROPERTY **Expectation of $U(1, n)$ — proof examinable**

If $X \sim U(1, n)$, then

$$E(X) = \frac{n+1}{2}.$$

Proof. By the definition of expectation for a discrete random variable,

$$E(X) = \sum_{r=1}^{n} r \cdot P(X = r) = \sum_{r=1}^{n} r \cdot \frac{1}{n} = \frac{1}{n} \sum_{r=1}^{n} r.$$

Using the first summation formula,

$$E(X) = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}. \quad \blacksquare$$

PROPERTY **Second Moment and Variance of $U(1, n)$ — proof examinable**

If $X \sim U(1, n)$, then

$$E(X^2) = \frac{(n+1)(2n+1)}{6}, \qquad \mathrm{Var}(X) = \frac{n^2 - 1}{12}.$$

Proof. First compute $E(X^2)$:

$$E(X^2) = \sum_{r=1}^{n} r^2 \cdot P(X = r) = \frac{1}{n} \sum_{r=1}^{n} r^2 = \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}.$$

Now use the computational formula $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$. Put both terms over the common denominator 12:

$$E(X^2) = \frac{(n+1)(2n+1)}{6} = \frac{2(n+1)(2n+1)}{12} = \frac{2(2n^2 + 3n + 1)}{12} = \frac{4n^2 + 6n + 2}{12},$$

$$[E(X)]^2 = \left(\frac{n+1}{2}\right)^2 = \frac{(n+1)^2}{4} = \frac{3(n+1)^2}{12} = \frac{3n^2 + 6n + 3}{12}.$$

Subtracting,

$$\mathrm{Var}(X) = \frac{(4n^2 + 6n + 2) - (3n^2 + 6n + 3)}{12} = \frac{n^2 - 1}{12}. \quad \blacksquare$$

WARNING **Algebraic Trap in the Variance Derivation**

When subtracting $\frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$, do not expand carelessly. The safest route is the common denominator 12 as shown above. A frequent error is to write $\frac{2n+1}{6} - \frac{n+1}{4} = \frac{4(2n+1)-3(n+1)}{12}$ and then mis-expand the numerator. Expand fully: $4(2n+1) = 8n+4$ and $3(n+1) = 3n+3$, giving $\frac{5n+1}{12}$ — which is wrong because the factor $(n+1)$ was pulled out incorrectly at the start. Work with the full numerators $4n^2+6n+2$ and $3n^2+6n+3$ instead; the $6n$ terms cancel and the result $\frac{n^2-1}{12}$ follows cleanly.

KEY POINTS **Summary for $X \sim U(1, n)$**

$$P(X = r) = \frac{1}{n} \quad (r = 1, \dots, n), \qquad E(X) = \frac{n+1}{2}, \qquad \mathrm{Var}(X) = \frac{n^2-1}{12}.$$

3.3 Extension: Uniform Distribution on $\{a, a+1, \dots, b\}$

In many practical situations the equally likely outcomes do not start at 1. If X takes the integer values $a, a+1, \dots, b$ with equal probability, where $a, b \in \mathbb{Z}$ and $a \le b$, then the number of outcomes is $n = b - a + 1$. We can write $X = (a-1) + Y$ where $Y \sim U(1, n)$. Using linearity of expectation, $E(X) = (a-1)+E(Y)$, and the fact that adding a constant does not change variance, $\mathrm{Var}(X) = \mathrm{Var}(Y)$. This gives the following general result.

PROPERTY **Uniform Distribution on $\{a, a+1, \dots, b\}$**

If X is uniformly distributed on $\{a, a+1, \dots, b\}$ (with $n = b - a + 1$ outcomes), then

$$P(X = r) = \frac{1}{n} \quad (r = a, \dots, b), \qquad E(X) = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{n^2-1}{12} = \frac{(b-a+1)^2-1}{12}.$$

Justification of the mean. $E(X) = (a-1) + \frac{n+1}{2} = \frac{2a-2+n+1}{2} = \frac{2a+(b-a+1)-1}{2} = \frac{a+b}{2}$. ■

EXEMPLE **Fair Die Revisited**

For a fair die, $a = 1$, $b = 6$, $n = 6$.

$$E(X) = \frac{1+6}{2} = 3.5, \qquad \mathrm{Var}(X) = \frac{6^2-1}{12} = \frac{35}{12} \approx 2.9167.$$

The standard deviation is $\sigma = \sqrt{35/12} \approx 1.708$.

3.4 Worked Examples

WORKED EXAMPLE 5 **Example 1: Expectation and Variance of a Fair Die Score**

A fair six-sided die is rolled once. Let X be the score obtained.

1. Write down the distribution of X.
2. Calculate $E(X)$ and $\mathrm{Var}(X)$.
3. If the die is rolled twice and the scores are added, find the mean and variance of the total score T.

SOLUTION

Solution.

1. $X \sim U(1, 6)$, so $P(X = r) = 1/6$ for $r = 1, \ldots, 6$.
2. Using the formulas with $n = 6$:

$$E(X) = \frac{6+1}{2} = 3.5, \qquad \mathrm{Var}(X) = \frac{6^2 - 1}{12} = \frac{35}{12}.$$

3. Let X_1, X_2 be the scores of the two independent rolls. Then $T = X_1 + X_2$. By linearity of expectation,

$$E(T) = E(X_1) + E(X_2) = 3.5 + 3.5 = 7.$$

Since the rolls are independent, variances add:

$$\mathrm{Var}(T) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) = \frac{35}{12} + \frac{35}{12} = \frac{35}{6} \approx 5.833.$$

WORKED EXAMPLE 6 **Example 2: Checking Fairness of a Game**

A game at a school fair uses a spinner with sectors numbered $2, 3, 4, 5, 6, 7$. A player pays 200 FCFA to spin once and wins 50 FCFA times the number shown. Determine whether the game is fair (i.e., expected net gain is zero) from the player's perspective.

SOLUTION

Solution. Let X be the number shown. X is uniform on $\{2, 3, 4, 5, 6, 7\}$, so $a = 2$, $b = 7$, $n = 6$.

$$E(X) = \frac{a+b}{2} = \frac{2+7}{2} = 4.5.$$

The winnings are $W = 50X$ (in FCFA). The cost to play is $C = 200$ FCFA. Net gain $G = W - C = 50X - 200$.

$$E(G) = 50E(X) - 200 = 50 \times 4.5 - 200 = 225 - 200 = 25 \text{ FCFA}.$$

Since $E(G) = 25 > 0$, the game is not fair — it favours the player. (A fair game would have $E(G) = 0$, which would require a cost of $50 \times 4.5 = 225$ FCFA per spin.)

3.5 Common Mistakes and Examination Tips

WARNING **Discrete vs. Continuous Uniform**

Do not confuse the discrete uniform distribution (this section) with the continuous uniform distribution (met later in the course).

- Discrete: X takes integer values $1, \ldots, n$ with $P(X = r) = 1/n$. Mean $= (n + 1)/2$, Variance $= (n^2 - 1)/12$.
- Continuous: X takes any real value in $[a, b]$ with density $f(x) = 1/(b - a)$. Mean $= (a + b)/2$, Variance $= (b - a)^2/12$.

The formulas look similar but the domains and the nature of probability (point masses vs. density) are completely different. In the GCE exam, the context (integer outcomes vs. continuous measurements) will tell you which one to use.

WARNING **Summation Limits and the Factor $1/n$**

When deriving $E(X)$ and $E(X^2)$, always write the summation from $r = 1$ to n explicitly, and do not forget the factor $1/n$ coming from $P(X = r) = 1/n$. A common error is to quote $E(X) = \sum r = n(n + 1)/2$ without dividing by n. Remember: $E(X) = \frac{1}{n}\sum_{r=1}^{n} r$.

WARNING **Variance Depends Only on the Number of Outcomes**

The variance $\frac{n^2-1}{12}$ is for the standard uniform on $\{1, \ldots, n\}$. If the distribution is on $\{a, \ldots, b\}$, the variance is still $\frac{n^2-1}{12}$ where $n = b - a + 1$ — it does not depend on a or b individually, only on the number of outcomes. Shifting the distribution does not change its spread. For example, $U(1, 6)$ and $U(2, 7)$ both have variance $35/12$.

3.6 Link with Frequency Distributions (Preview)

In the lesson on the relationship between probability distributions and frequency distributions, we will see how a theoretical probability distribution like $U(1, n)$ can be compared with an observed frequency distribution. For instance, if we roll a die 600 times and record the frequency of each face, we expect each face to appear about $600 \times \frac{1}{6} = 100$ times. These expected frequencies $E_r = N \cdot P(X = r)$ can then be compared with the observed frequencies to judge whether

the die is truly fair — a comparison formalised by the χ^2 goodness-of-fit test met in the statistics course.

EXERCISE 3 **Practice Exercises**

1. A random number generator produces integers uniformly from 10 to 20 inclusive. Find the mean and variance of the generated number.
2. A bag contains tickets numbered 1 to 50. One ticket is drawn at random. Let X be the number on the ticket. Find $P(X > 40)$, $E(X)$ and $\mathrm{Var}(X)$.
3. A game uses a spinner with sectors $1, 2, 3, 4, 5$. It costs 100 FCFA to play; you win 30 FCFA times the number shown. Is the game fair? If not, who does it favour and by how much on average?
4. Prove that for $X \sim U(1, n)$, $E(X^2) = \dfrac{(n+1)(2n+1)}{6}$, and deduce that $\mathrm{Var}(X) = \dfrac{n^2 - 1}{12}$.
5. A fair die is rolled 360 times. Write down the expected frequency of each face. What is the expected total of all 360 scores?

SOLUTION **Exercise 1**

X is uniform on $\{10, 11, \ldots, 20\}$. Here $a = 10$, $b = 20$, $n = 20 - 10 + 1 = 11$. $E(X) = \dfrac{10 + 20}{2} = 15$. $\mathrm{Var}(X) = \dfrac{11^2 - 1}{12} = \dfrac{120}{12} = 10$.

SOLUTION **Exercise 2**

$X \sim U(1, 50)$. $P(X > 40) = P(X = 41, \ldots, 50) = 10 \times \dfrac{1}{50} = 0.2$. $E(X) = \dfrac{50 + 1}{2} = 25.5$. $\mathrm{Var}(X) = \dfrac{50^2 - 1}{12} = \dfrac{2499}{12} = 208.25$.

SOLUTION **Exercise 3**

$X \sim U(1, 5)$, so $E(X) = \dfrac{5 + 1}{2} = 3$. Winnings $W = 30X$, cost $C = 100$ FCFA. Net gain $G = 30X - 100$. $E(G) = 30E(X) - 100 = 30 \times 3 - 100 = -10$ FCFA. The game is not fair; it favours the organiser by 10 FCFA per play on average.

SOLUTION **Exercise 4**

$E(X^2) = \sum_{r=1}^{n} r^2 P(X = r) = \frac{1}{n}\sum_{r=1}^{n} r^2 = \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}$. Then

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{4n^2+6n+2}{12} - \frac{3n^2+6n+3}{12} = \frac{n^2-1}{12}.$$

SOLUTION **Exercise 5**

Expected frequency of each face $= 360 \times \frac{1}{6} = 60$. Expected score per roll $= 3.5$, so the expected total of all 360 scores is $360 \times 3.5 = 1260$.

4 THE GEOMETRIC DISTRIBUTION

In the previous section we studied the discrete uniform distribution, where every outcome in a finite set is equally likely. We now turn to a distribution that models a very different kind of random experiment: we repeat independent trials, each with the same probability of success, until the first success occurs. The random variable is the number of trials needed to obtain that first success. This is the **geometric distribution**, a cornerstone for modelling waiting times in discrete settings.

4.1 Definition and Recognition of Geometric Situations

EXPERIMENT **A Motivating Situation**

A trader in Douala sells phone credit. Experience shows that each passer-by she approaches buys credit with probability 0.2, independently of the others. She wonders: how many people must she approach before she makes her first sale? It could be the very first person (probability 0.2), or the second (a failure then a success: 0.8×0.2), or the third ($0.8 \times 0.8 \times 0.2$), and so on. The number of approaches needed is a random variable whose probabilities form a geometric sequence: this is exactly the situation modelled by the geometric distribution.

DEFINITION **Geometric Distribution**

Let an experiment consist of independent Bernoulli trials, each with probability of success p ($0 < p < 1$) and probability of failure $q = 1 - p$. Let X be the random variable represent-

ing the number of trials up to and including the first success. Then X follows a **geometric distribution** with parameter p, denoted $X \sim \text{Geo}(p)$. Its probability mass function is

$$P(X = r) = q^{r-1}p \qquad (r = 1, 2, 3, \ldots)$$

The formula is immediate: the event $\{X = r\}$ means that the first $r-1$ trials are failures and the r-th trial is a success. By independence,

$$P(X = r) = \underbrace{q \times q \times \cdots \times q}_{r-1 \text{ failures}} \times p = q^{r-1}p.$$

KEY POINTS **Recognising a Geometric Setting**

A random variable X has a geometric distribution if the situation satisfies:

- Trials are independent.
- Each trial has only two outcomes: success (probability p) and failure (probability $q = 1 - p$).
- The probability p remains constant from trial to trial.
- We count the number of trials until the first success occurs (including that success).

Typical examples: the number of times you roll a die until you get a six; the number of job interviews until the first offer; the number of items inspected until the first defective one is found.

WARNING **Two Common Definitions — Check the Convention!**

Some textbooks (and some software packages) define the geometric variable as the number of failures before the first success, taking values $0, 1, 2, \ldots$ with $P(Y = k) = q^k p$. The GCE Advanced Level convention uses the definition above: X is the number of trials including the first success, taking values $1, 2, 3, \ldots$ with $P(X = r) = q^{r-1}p$. Using the wrong convention shifts the mean by 1 and changes the variance formula, so always state clearly which variable you are counting.

EXEMPLE **Identifying a Geometric Variable**

A factory produces light bulbs; 5% are defective. Bulbs are tested one by one until the first defective bulb is found. Let X be the number of bulbs tested.

- Trials: testing a bulb. Independent? Yes (large batch).
- Success: finding a defective bulb, $p = 0.05$, $q = 0.95$.

- X counts trials up to and including the first defective, so $X \sim \text{Geo}(0.05)$.
- $P(X = 3) = (0.95)^2 \times 0.05 = 0.045125$.

4.2 Cumulative Probabilities and the Tail Formula

Often we need probabilities such as "more than r trials are needed" or "at most r trials". Summing the probability mass function term by term is possible but tedious. A simple closed form exists for the upper tail.

PROPERTY **Tail Probability** $P(X > r)$

For $X \sim \text{Geo}(p)$,

$$P(X > r) = q^r \qquad (r = 0, 1, 2, \dots)$$

Proof: The event $\{X > r\}$ means that the first r trials are all failures. By independence, its probability is $q \times q \times \cdots \times q = q^r$. □

From this we immediately obtain:

$$P(X \le r) = 1 - q^r, \qquad P(X \ge r) = q^{r-1}, \qquad P(X < r) = 1 - q^{r-1}.$$

WARNING **Strict or Non-Strict Inequalities**

A very common examination error is to confuse $P(X > r) = q^r$ with $P(X \ge r) = q^{r-1}$. Remember: $P(X > r)$ requires r failures, while $P(X \ge r)$ requires only $r - 1$ failures (the r-th trial is free). Always translate the sentence into an event on the trials before applying the formula.

EXEMPLE **Using the Tail Formula**

With $p = 0.05$ ($q = 0.95$) as above, find the probability that more than 10 bulbs are tested before the first defective appears.

$$P(X > 10) = (0.95)^{10} \approx 0.5987.$$

Hence $P(X \le 10) = 1 - 0.5987 = 0.4013$.

4.3 Expectation and Variance

The mean and variance of the geometric distribution have elegant closed forms. Their derivations use standard power-series manipulations and are examinable at this level.

PROPERTY **Mean and Variance of** Geo(p)

If $X \sim \text{Geo}(p)$ with $q = 1 - p$, then

$$E(X) = \frac{1}{p}, \qquad \text{Var}(X) = \frac{q}{p^2}.$$

Proof (examinable). We derive both results from the geometric series and its derivatives. Step 1: the mean.

$$E(X) = \sum_{r=1}^{\infty} r\, q^{r-1} p = p \sum_{r=1}^{\infty} r\, q^{r-1}.$$

Recall the geometric series, valid for $|q| < 1$:

$$\sum_{r=0}^{\infty} q^r = \frac{1}{1-q} = \frac{1}{p}.$$

Differentiate both sides with respect to q:

$$\sum_{r=1}^{\infty} r\, q^{r-1} = \frac{1}{(1-q)^2} = \frac{1}{p^2}.$$

Hence

$$E(X) = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$

Step 2: the variance. We first compute $E[X(X-1)]$, which avoids subtracting large nearly equal numbers:

$$E[X(X-1)] = \sum_{r=2}^{\infty} r(r-1)\, q^{r-1} p = pq \sum_{r=2}^{\infty} r(r-1)\, q^{r-2}.$$

Differentiate the identity $\sum_{r=1}^{\infty} r\, q^{r-1} = \frac{1}{(1-q)^2}$ once more with respect to q:

$$\sum_{r=2}^{\infty} r(r-1)\, q^{r-2} = \frac{2}{(1-q)^3} = \frac{2}{p^3}.$$

Therefore

$$E[X(X-1)] = pq \cdot \frac{2}{p^3} = \frac{2q}{p^2}.$$

Now $E(X^2) = E[X(X-1)] + E(X) = \frac{2q}{p^2} + \frac{1}{p} = \frac{2q+p}{p^2} = \frac{1+q}{p^2}$, since $2q + p = q + (q+p) = 1 + q$. Finally,

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1+q}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}.$$

□

EXEMPLE **Mean and Variance in Context**

A basketball player has a free-throw success rate of $p = 0.7$. Let X be the number of shots until the first basket.

- $E(X) = 1/0.7 \approx 1.43$ shots.
- $\mathrm{Var}(X) = 0.3/(0.7)^2 = 0.3/0.49 \approx 0.612$.
- Standard deviation $\sigma \approx 0.78$ shots.

On average the player needs about 1.4 shots to score; the small standard deviation reflects that success usually comes quickly when p is large.

4.4 The Mode of the Geometric Distribution

PROPERTY **Mode of** Geo(p)

For $X \sim \mathrm{Geo}(p)$ with $0 < p < 1$, the mode is always 1. Proof: $P(X = r) = q^{r-1}p$. Since $0 < q < 1$, the sequence is strictly decreasing:

$$\frac{P(X = r+1)}{P(X = r)} = q < 1 \quad \text{for all } r \geq 1,$$

so $P(X = 1) = p$ is the largest probability. □

This matches intuition: the single most likely scenario is that the very first trial is a success, whatever the value of p.

4.5 The Memoryless Property

The geometric distribution is the only discrete distribution with the **memoryless property**: past failures do not affect the distribution of the remaining waiting time.

PROPERTY **Memoryless Property**

For $X \sim \mathrm{Geo}(p)$ and any non-negative integers m, n,

$$P(X > m + n \mid X > m) = P(X > n).$$

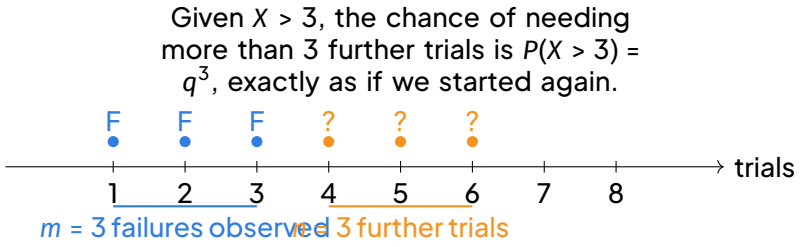
Proof: Since $\{X > m + n\} \subset \{X > m\}$,

$$P(X > m + n \mid X > m) = \frac{P(X > m + n)}{P(X > m)} = \frac{q^{m+n}}{q^m} = q^n = P(X > n).$$

□

DID YOU KNOW? **Interpretation**

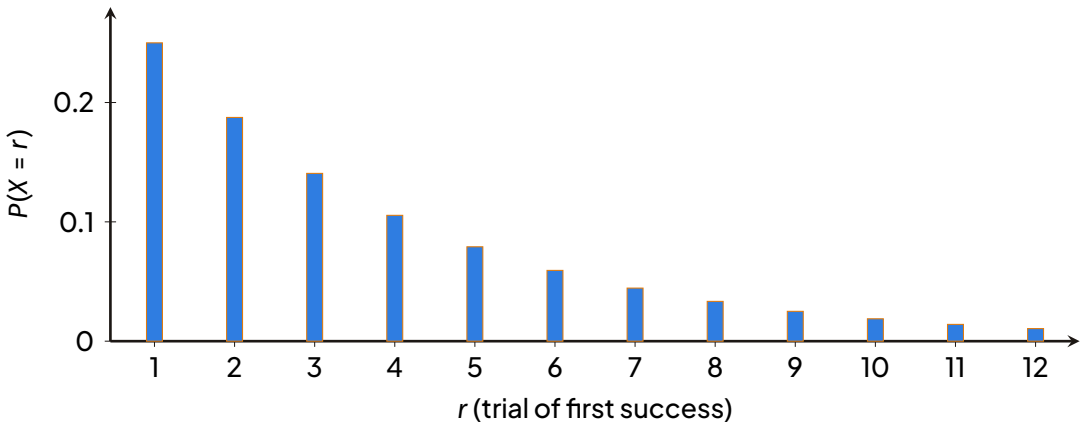
If you have already observed m failures, the conditional distribution of the additional number of trials needed for the first success is again Geo(p). The process "restarts" with no memory of the past. This is why the geometric distribution models systems that do not age, such as repeated independent attempts or certain queueing situations.



The memoryless property: after m observed failures, the remaining waiting time has the same geometric distribution as at the start.

4.6 Visualising the Geometric Distribution

The probability mass function decreases geometrically: each bar is q times the previous one. The following bar chart shows $X \sim \text{Geo}(0.25)$ for $r = 1, \ldots, 12$.



Bar chart of $X \sim \text{Geo}(0.25)$ for $r = 1$ to 12. The probabilities decay by a factor of $q = 0.75$ at each step, and the mode is 1.

4.7 Applied Problems and Worked Examples

We now solve typical examination questions that combine recognition, tail probabilities, expectation, variance, and the memoryless property.

WORKED EXAMPLE 7 **Finding p from a Given Probability**

The random variable $X \sim \text{Geo}(p)$. It is given that $P(X > 4) = 0.2401$. Find the value of p, and hence calculate $E(X)$ and $\text{Var}(X)$.

SOLUTION

Solution: Step 1: Use the tail formula $P(X > r) = q^r$ with $r = 4$:

$$q^4 = 0.2401.$$

Step 2: Take the fourth root. Since $0.7^2 = 0.49$ and $0.49^2 = 0.2401$, we get

$$q = 0.7, \qquad p = 1 - q = 0.3.$$

Step 3: Compute the mean and variance:

$$E(X) = \frac{1}{p} = \frac{1}{0.3} = \frac{10}{3} \approx 3.33, \qquad \text{Var}(X) = \frac{q}{p^2} = \frac{0.7}{0.09} = \frac{70}{9} \approx 7.78.$$

WORKED EXAMPLE 8 **Memoryless Property in Action**

A machine produces components; 2% are defective. Components are tested one by one until the first defective is found. Given that the first 20 components tested are all non-defective, find the probability that more than 15 further components need to be tested.

SOLUTION

Solution: Let $X \sim \text{Geo}(0.02)$, so $q = 0.98$. The condition "the first 20 are non-defective" means $X > 20$. We require

$$P(X > 20 + 15 \mid X > 20) = P(X > 35 \mid X > 20).$$

By the memoryless property this equals

$$P(X > 15) = q^{15} = (0.98)^{15} \approx 0.7386.$$

So the probability is approximately 0.739: the twenty good components already tested give no information about the future.

WORKED EXAMPLE 9 **Expected Cost of Testing**

In a production line, each item tested costs 500 CFA francs. The probability that an item is defective is 0.04. Testing stops at the first defective item. Find the expected total cost and the standard deviation of the cost.

SOLUTION

Solution: Let $X \sim \text{Geo}(0.04)$ be the number of items tested; the cost is $C = 500X$ CFA.

$$E(X) = \frac{1}{0.04} = 25, \qquad \text{Var}(X) = \frac{0.96}{(0.04)^2} = \frac{0.96}{0.0016} = 600.$$

Using the linearity of expectation and $\text{Var}(aX) = a^2\text{Var}(X)$:

$$E(C) = 500 \times 25 = 12\,500 \text{ CFA},$$

$$\text{Var}(C) = 500^2 \times 600 = 150\,000\,000 \text{ CFA}^2, \qquad \sigma_C = \sqrt{150\,000\,000} \approx 12\,247 \text{ CFA}.$$

WORKED EXAMPLE 10 **A Two-Stage Problem**

A fair coin is tossed repeatedly. Find the probability that the first head appears on an even-numbered toss.

SOLUTION

Solution: Let $X \sim \text{Geo}(0.5)$ be the toss of the first head, with $p = q = \frac{1}{2}$. We need

$$P(X \text{ even}) = P(X = 2) + P(X = 4) + P(X = 6) + \cdots = qp + q^3p + q^5p + \cdots$$

This is a geometric series with first term $qp = \frac{1}{4}$ and common ratio $q^2 = \frac{1}{4}$:

$$P(X \text{ even}) = \frac{qp}{1 - q^2} = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}.$$

Check: by symmetry of the argument, $P(X \text{ odd}) = \dfrac{p}{1 - q^2} = \dfrac{2}{3}$, and indeed $\frac{1}{3} + \frac{2}{3} = 1$.

KEY POINTS **Key Formulas for** Geo(p)

- PMF: $P(X = r) = q^{r-1}p, r = 1, 2, 3, \ldots$
- Tail: $P(X > r) = q^r, \quad P(X \le r) = 1 - q^r, \quad P(X \ge r) = q^{r-1}.$
- Mean: $E(X) = \dfrac{1}{p}.$
- Variance: $\text{Var}(X) = \dfrac{q}{p^2}.$
- Mode: 1 (always).

- Memoryless: $P(X > m + n \mid X > m) = P(X > n) = q^n$.

METHOD **Solving Geometric Distribution Problems**

1. Identify the success event and its probability p. Confirm independence and constant p.
2. Define X = number of trials up to and including the first success. State $X \sim \text{Geo}(p)$.
3. Use the tail formula $P(X > r) = q^r$ for "more than r trials"; avoid summing long series, and watch strict versus non-strict inequalities.
4. Apply $E(X) = 1/p$ and $\text{Var}(X) = q/p^2$ when expectations or variances are needed.
5. Check whether the memoryless property applies when the problem conditions on past failures.
6. Interpret the result in the context of the problem (expected cost, number of attempts, etc.).

EXERCISE 4 **Practice Problems**

1. A fair die is rolled until a six appears. Find (a) the probability that exactly 4 rolls are needed, (b) the probability that more than 5 rolls are needed, (c) the expected number of rolls and the variance.
2. In a certain region, the probability that a randomly selected household has a solar panel is 0.12. Households are visited until one with a solar panel is found. Find the probability that at most 3 households are visited. Given that the first 5 households visited have no solar panel, find the probability that more than 4 further households must be visited.
3. A random variable $X \sim \text{Geo}(p)$ has variance 6. Find p and $E(X)$.
4. Each trial costs 2000 CFA francs. If $X \sim \text{Geo}(0.2)$ is the number of trials until the first success, find the expected cost and the standard deviation of the total cost.
5. Show that for $X \sim \text{Geo}(p)$, $P(X \text{ is even}) = \dfrac{q}{1+q}$, and deduce the value of this probability when $p = 0.4$.

SOLUTION **Exercise 1**

$p = 1/6$, $q = 5/6$. **(a)** $P(X = 4) = q^3 p = (5/6)^3(1/6) = 125/1296 \approx 0.0965$. **(b)** $P(X > 5) = q^5 = (5/6)^5 = 3125/7776 \approx 0.4019$. **(c)** $E(X) = 1/p = 6$; $\mathrm{Var}(X) = q/p^2 = (5/6)/(1/36) = 30$.

SOLUTION **Exercise 2**

$p = 0.12$, $q = 0.88$. $P(X \le 3) = 1 - q^3 = 1 - (0.88)^3 = 1 - 0.681472 = 0.318528 \approx 0.319$. Given $X > 5$, the memoryless property gives

$$P(X > 5 + 4 \mid X > 5) = P(X > 4) = q^4 = (0.88)^4 \approx 0.5997.$$

SOLUTION **Exercise 3**

$\mathrm{Var}(X) = q/p^2 = 6$ with $q = 1 - p$ gives $1 - p = 6p^2$, i.e. $6p^2 + p - 1 = 0$.

$$p = \frac{-1 + \sqrt{1 + 24}}{12} = \frac{-1 + 5}{12} = \frac{1}{3}$$

(taking the positive root, since $0 < p < 1$). Hence $E(X) = 1/p = 3$.

SOLUTION **Exercise 4**

$X \sim \mathrm{Geo}(0.2)$: $E(X) = 1/0.2 = 5$, $\mathrm{Var}(X) = 0.8/(0.2)^2 = 0.8/0.04 = 20$. Cost $C = 2000X$: $E(C) = 2000 \times 5 = 10\,000$ CFA; $\mathrm{Var}(C) = 2000^2 \times 20 = 80\,000\,000$ CFA^2; $\sigma_C = \sqrt{80\,000\,000} \approx 8944$ CFA.

SOLUTION **Exercise 5**

$P(X \text{ even}) = qp + q^3 p + q^5 p + \cdots = \dfrac{qp}{1 - q^2} = \dfrac{qp}{(1 - q)(1 + q)} = \dfrac{q}{1 + q}$, since $p = 1 - q$. For $p = 0.4$, $q = 0.6$: $P(X \text{ even}) = \dfrac{0.6}{1.6} = \dfrac{3}{8} = 0.375$.

This completes our study of the geometric distribution. In the next section we will explore the Poisson distribution, which models the number of events occurring in a fixed interval of time or space, and its intimate connection with the binomial distribution.

5 THE POISSON DISTRIBUTION AND ITS TABLES

In the previous section we studied the **geometric distribution**, which counts the number of trials needed to obtain the first success. In many real situations, however, there is no fixed sequence of trials at all: phone calls arrive at a call centre in Douala at unpredictable instants, vehicles pass a checkpoint on the Yaoundé–Bafoussam road at random times, and typing errors appear scattered through the pages of a manuscript. What we can count is the number of events occurring in a fixed interval of time or space. The distribution that models exactly this situation is the **Poisson distribution**, named after the French mathematician Siméon Denis Poisson. It is one of the most frequently examined distributions at the GCE Advanced Level.

5.1 Definition and Conditions for a Poisson Process

EXPERIMENT **Counting random events**

Suppose a small clinic in Bamenda receives, on average, 3 emergency patients per hour. If we observe the arrivals during one hour, we might record $0, 1, 2, 3, \ldots$ patients. The arrivals are not scheduled: they occur at random, independently of one another, but the average rate of 3 per hour stays roughly constant throughout the day. Any count of events satisfying these three features is a candidate for a Poisson model.

DEFINITION **Poisson distribution**

A discrete random variable X follows a **Poisson distribution** with parameter $\lambda > 0$, written $X \sim \text{Po}(\lambda)$, if

$$P(X = r) = e^{-\lambda}\frac{\lambda^r}{r!}, \qquad r = 0, 1, 2, 3, \ldots$$

X models the number of events occurring in a fixed interval (of time, length, area or volume) when the events occur:

- randomly (no predictable pattern);
- independently (one event does not influence the next);
- at a constant average rate λ per unit interval;
- singly (two events cannot occur at exactly the same instant).

The parameter λ is the mean number of events per unit interval.

WARNING **When the Poisson model fails**

Do not use a Poisson model if events arrive in clusters (e.g. students leaving school in groups at closing time: not independent), or if the rate changes over the interval (traf-

fic at 7 a.m. versus midnight: rate not constant). Always check the four conditions before writing $X \sim \text{Po}(\lambda)$.

PROPERTY **The Poisson probabilities form a genuine distribution**

The probabilities $P(X = r) = e^{-\lambda}\lambda^r / r!$ are non-negative and sum to 1. The proof is examinable.

Proof. Using the exponential series $e^x = \sum_{r=0}^{\infty} \frac{x^r}{r!}$, valid for every real x:

$$\sum_{r=0}^{\infty} P(X = r) = \sum_{r=0}^{\infty} e^{-\lambda}\frac{\lambda^r}{r!} = e^{-\lambda}\sum_{r=0}^{\infty}\frac{\lambda^r}{r!} = e^{-\lambda}e^{\lambda} = e^{0} = 1.$$

Since also $e^{-\lambda} > 0$ and $\lambda^r / r! \geq 0$, every probability is non-negative. ■

5.2 Mean and Variance of the Poisson Distribution

The most remarkable feature of the Poisson distribution is that its mean and variance are equal.

PROPERTY **Mean and variance**

If $X \sim \text{Po}(\lambda)$, then

$$E(X) = \lambda \qquad \text{and} \qquad \text{Var}(X) = \lambda.$$

The equality $E(X) = \text{Var}(X)$ is the signature of the Poisson distribution. The proof is examinable.

Proof of $E(X) = \lambda$.

$$E(X) = \sum_{r=0}^{\infty} r\, e^{-\lambda}\frac{\lambda^r}{r!} = e^{-\lambda}\sum_{r=1}^{\infty}\frac{\lambda^r}{(r-1)!} \quad \text{(the } r = 0 \text{ term vanishes)}$$

$$= e^{-\lambda}\lambda\sum_{r=1}^{\infty}\frac{\lambda^{r-1}}{(r-1)!} = \lambda\, e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = \lambda\, e^{-\lambda}e^{\lambda} = \lambda.$$

Proof of $\text{Var}(X) = \lambda$ via $E\big(X(X-1)\big)$. We use the identity $\text{Var}(X) = E\big(X(X-1)\big) + E(X) - \big(E(X)\big)^2$, which follows from $X^2 = X(X-1) + X$. First:

$$E\big(X(X-1)\big) = \sum_{r=0}^{\infty} r(r-1)\, e^{-\lambda}\frac{\lambda^r}{r!} = e^{-\lambda}\sum_{r=2}^{\infty}\frac{\lambda^r}{(r-2)!}$$

$$= \lambda^2 e^{-\lambda}\sum_{r=2}^{\infty}\frac{\lambda^{r-2}}{(r-2)!} = \lambda^2 e^{-\lambda}e^{\lambda} = \lambda^2.$$

Therefore $\mathrm{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$. ■

METHOD **Quick check of a Poisson model from data**

1. Compute the sample mean $\bar{x}$ and the sample variance s^2 of the counts.
2. If $\bar{x} \approx s^2$, a Poisson model is plausible; take $\lambda \approx \bar{x}$.
3. If s^2 is very different from $\bar{x}$, reject the Poisson model.

5.3 The Recurrence Formula and the Mode

Computing $P(X = r)$ directly involves large powers and factorials. The following recurrence makes successive probabilities easy to obtain and also locates the **mode**.

PROPERTY **Recurrence formula for the Poisson distribution**

If $X \sim \mathrm{Po}(\lambda)$, then

$$\frac{P(X = r+1)}{P(X = r)} = \frac{\lambda}{r+1}, \qquad \text{i.e.} \qquad P(X = r+1) = \frac{\lambda}{r+1}\,P(X = r).$$

Consequently the probabilities increase while $r + 1 < \lambda$ and decrease once $r + 1 > \lambda$: the mode is the integer part of λ (and if λ is itself an integer, both $\lambda - 1$ and λ are modes, with equal probabilities).

Proof.

$$\frac{P(X = r+1)}{P(X = r)} = \frac{e^{-\lambda}\lambda^{r+1}/(r+1)!}{e^{-\lambda}\lambda^{r}/r!} = \frac{\lambda}{r+1}.$$

The ratio exceeds 1 exactly when $r + 1 < \lambda$, which gives the statement on the mode. ■

EXEMPLE **Using the recurrence to find the mode**

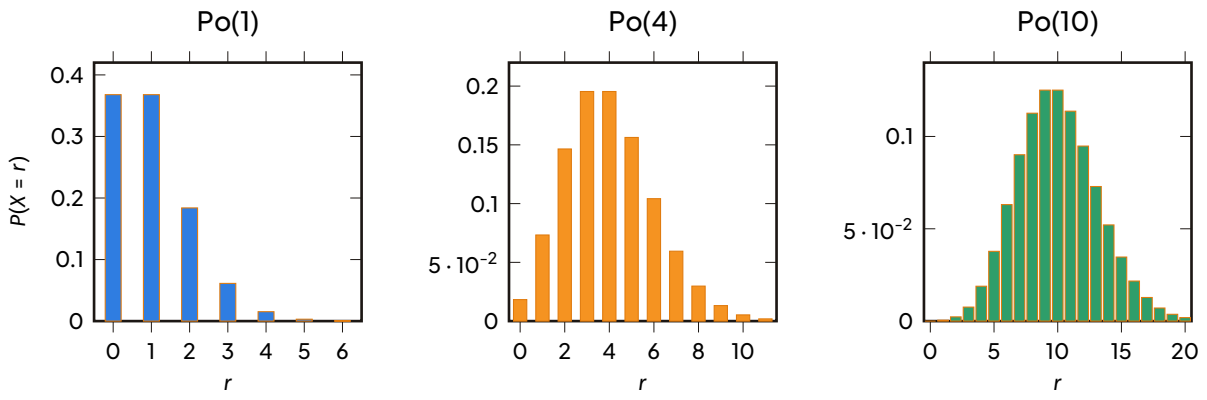
Let $X \sim \mathrm{Po}(4)$. Starting from $P(X = 0) = e^{-4} \approx 0.0183$:

$$P(X = 1) = \tfrac{4}{1}P(X = 0) \approx 0.0733, \qquad P(X = 2) = \tfrac{4}{2}P(X = 1) \approx 0.1465,$$
$$P(X = 3) = \tfrac{4}{3}P(X = 2) \approx 0.1954, \qquad P(X = 4) = \tfrac{4}{4}P(X = 3) \approx 0.1954,$$
$$P(X = 5) = \tfrac{4}{5}P(X = 4) \approx 0.1563.$$

The probabilities stop increasing at $r = 4$; since $\lambda = 4$ is an integer, $P(X = 3) = P(X = 4)$ and the distribution has the two modes 3 and 4.

5.4 The Shape of the Distribution

The bar charts below show Po(1), Po(4) and Po(10). For small λ the distribution is strongly skewed to the right; as λ grows it becomes more symmetric and bell-shaped.



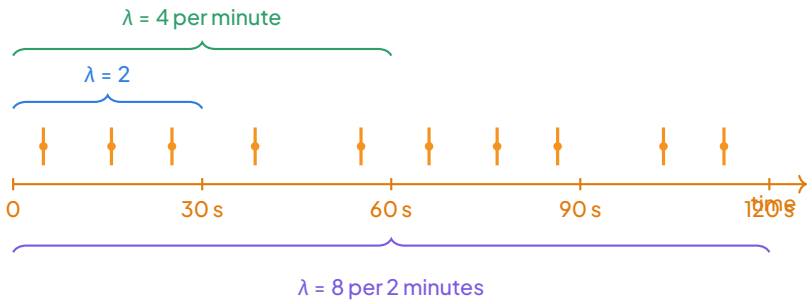
Bar charts of Po(1), Po(4) and Po(10): the distribution becomes more symmetric as λ increases.

5.5 The Unit Interval: Rescaling λ

The parameter λ is always attached to a particular interval. If the question changes the interval, you must rescale λ in proportion, because the average rate is constant.

METHOD **Rescaling the parameter**

1. Identify the rate and its interval (e.g. 4 calls per minute).
2. Identify the interval asked in the question (e.g. 30 seconds = $\frac{1}{2}$ minute).
3. Rescale proportionally: $\lambda_{\text{new}} = 4 \times \frac{1}{2} = 2$ calls per 30 seconds.
4. Use $X \sim \text{Po}(\lambda_{\text{new}})$ for all calculations.



Random events (orange ticks) on a timeline: the rate is constant, so λ scales with the length of the interval.

WORKED EXAMPLE 11 **Rescaling and "at least one"**

Calls arrive at an office in Yaoundé at random, independently, at an average rate of 4 per minute. Find the probability that (a) exactly 2 calls arrive in a given 30-second period; (b) at least one call arrives in a given 30-second period.

SOLUTION

Solution. Rescale: $\lambda = 4 \times \frac{30}{60} = 2$ per 30 seconds, so $X \sim \text{Po}(2)$.

(a) $P(X = 2) = e^{-2}\frac{2^2}{2!} = 2e^{-2} \approx 0.2707$.

(b) $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-2} \approx 1 - 0.1353 = 0.8647$.

WARNING **Exam tip: the fastest route for "at least one"**

For $X \sim \text{Po}(\lambda)$,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-\lambda}.$$

Never sum $P(X = 1) + P(X = 2) + \ldots$: the series is infinite. The complement is one line of work.

5.6 Using Cumulative Poisson Tables

Cumulative Poisson tables give $P(X \leq r)$ for selected values of λ and r. Every other probability is obtained from these:

KEY POINTS **Reading cumulative Poisson tables**

For $X \sim \text{Po}(\lambda)$, with $F(r) = P(X \leq r)$ read from the table:

- $P(X = r) = P(X \leq r) - P(X \leq r - 1)$;

- $P(X \geq r) = 1 - P(X \leq r-1)$;
- $P(X > r) = 1 - P(X \leq r)$;
- $P(a \leq X \leq b) = P(X \leq b) - P(X \leq a-1)$.

If λ is not tabulated, compute $e^{-\lambda}\lambda^r / r!$ directly with a calculator.

EXEMPLE **Worked table reading**

Let $X \sim \text{Po}(3.5)$. From cumulative tables: $P(X \leq 1) = 0.1359$, $P(X \leq 2) = 0.3208$, $P(X \leq 3) = 0.5366$, $P(X \leq 5) = 0.8576$.

- $P(X = 3) = P(X \leq 3) - P(X \leq 2) = 0.5366 - 0.3208 = 0.2158$;
- $P(X \geq 4) = 1 - P(X \leq 3) = 1 - 0.5366 = 0.4634$;
- $P(2 \leq X \leq 5) = P(X \leq 5) - P(X \leq 1) = 0.8576 - 0.1359 = 0.7217$.

WARNING **Off-by-one errors**

The most common table mistake is confusing $P(X \geq r) = 1 - P(X \leq r-1)$ with $1 - P(X \leq r)$. Remember: the Poisson variable is discrete, so $P(X \geq r)$ includes r, and its complement is $P(X \leq r-1)$.

5.7 Sum of Two Independent Poisson Variables

PROPERTY **Additivity of the Poisson distribution**

If $X \sim \text{Po}(\lambda)$ and $Y \sim \text{Po}(\mu)$ are independent, then

$$X + Y \sim \text{Po}(\lambda + \mu).$$

The proof is instructive and may be required.

Proof. Let $Z = X + Y$. For $n \geq 0$, the event $\{Z = n\}$ is the disjoint union of the events $\{X = r\} \cap \{Y = n - r\}$ for $r = 0, 1, \ldots, n$. Using independence and then the binomial theorem:

$$P(Z = n) = \sum_{r=0}^{n} P(X = r)\,P(Y = n - r) = \sum_{r=0}^{n} e^{-\lambda}\frac{\lambda^r}{r!}\,e^{-\mu}\frac{\mu^{n-r}}{(n-r)!}$$

$$= e^{-(\lambda+\mu)}\frac{1}{n!}\sum_{r=0}^{n}\frac{n!}{r!(n-r)!}\lambda^r\mu^{n-r} = e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^n}{n!},$$

since $\sum_{r=0}^{n} \binom{n}{r} \lambda^r \mu^{n-r} = (\lambda + \mu)^n$. This is exactly the Po($\lambda + \mu$) formula. ■

Intuitively this is natural: if accidents occur at junction A at an average rate of 2 per week and independently at junction B at a rate of 1.5 per week, then the combined stream of accidents still occurs randomly, independently and at the constant combined rate 3.5 per week.

WARNING **The difference is NOT Poisson**

Only the sum of independent Poisson variables is Poisson. The difference $X - Y$ takes negative values, so it cannot be Poisson (a Poisson variable only takes values $0, 1, 2, \ldots$). For $X - Y$ you must work from first principles, e.g. $P(X > Y) = \sum_r P(Y = r)\, P(X \geq r + 1)$.

WORKED EXAMPLE 12 **Accidents at two junctions**

Accidents occur independently at two junctions of Douala: at junction A at an average rate of 2 per week, and at junction B at an average rate of 1.5 per week. Find the probability that, in a given week, (a) there are exactly 4 accidents in total at the two junctions; (b) there is at least one accident in total.

SOLUTION

Solution. Let $X \sim \text{Po}(2)$ and $Y \sim \text{Po}(1.5)$, independent. Then $T = X + Y \sim \text{Po}(3.5)$.

(a) $P(T = 4) = e^{-3.5}\dfrac{3.5^4}{4!} = e^{-3.5}\dfrac{150.0625}{24} \approx 0.0302 \times 6.2526 \approx 0.1888.$

(b) $P(T \geq 1) = 1 - e^{-3.5} \approx 1 - 0.0302 = 0.9698.$

EXERCISE 5 **Consolidation**

1. Bacteria occur randomly in a liquid at an average rate of 5 per millilitre. A sample of 0.4 ml is examined. Find the probability that it contains (a) exactly 3 bacteria; (b) at least one bacterium.
2. Given $X \sim \text{Po}(6)$ and cumulative values $P(X \leq 4) = 0.2851$, $P(X \leq 8) = 0.8472$, find $P(5 \leq X \leq 8)$ and $P(X \geq 9)$.
3. $X \sim \text{Po}(2.4)$ and $Y \sim \text{Po}(3.1)$ are independent. State the distribution of $X+Y$ and compute $P(X + Y = 0)$ and the common value of $E(X + Y)$ and $\text{Var}(X + Y)$.
4. For $X \sim \text{Po}(2.5)$, use the recurrence formula to find the mode of the distribution.

SOLUTION **Exercise 1**

1. Rescale: $\lambda = 5 \times 0.4 = 2$, so $X \sim \text{Po}(2)$. **(a)** $P(X = 3) = e^{-2}\frac{2^3}{3!} = \frac{4}{3}e^{-2} \approx 0.1804$. **(b)** $P(X \geq 1) = 1 - e^{-2} \approx 0.8647$.
2. $P(5 \leq X \leq 8) = P(X \leq 8) - P(X \leq 4) = 0.8472 - 0.2851 = 0.5621$; $P(X \geq 9) = 1 - P(X \leq 8) = 0.1528$.
3. $X + Y \sim \text{Po}(5.5)$; $P(X + Y = 0) = e^{-5.5} \approx 0.0041$; $E(X + Y) = \text{Var}(X + Y) = 5.5$.
4. The probabilities increase while $r + 1 < 2.5$, i.e. for $r = 0, 1$, and decrease from $r + 1 = 3$ onwards. Hence the mode is $r = 2$: indeed $P(X = 2) = e^{-2.5}\frac{2.5^2}{2!} \approx 0.2565$, larger than $P(X = 1) \approx 0.2052$ and $P(X = 3) \approx 0.2138$.

DID YOU KNOW? **Poisson and the Prussian army**

Poisson introduced this law in 1837 in a work on the probability of judgments in criminal and civil matters. It became famous through Bortkiewicz's 1898 study of the number of Prussian soldiers killed each year by horse kicks: the yearly counts matched a Poisson distribution remarkably well. Today the same law models calls to mobile networks, arrivals at bank counters in Douala, and radioactive decays.

KEY POINTS **Key points of the section**

- $X \sim \text{Po}(\lambda)$: $P(X = r) = e^{-\lambda}\lambda^r / r!$, for events random, independent, at constant rate λ.
- $E(X) = \text{Var}(X) = \lambda$: equal mean and variance is the Poisson signature.
- Recurrence: $P(X = r + 1) = \frac{\lambda}{r + 1}P(X = r)$; the mode is the integer part of λ ($\lambda - 1$ and λ if λ is an integer).
- Always rescale λ to the interval asked before computing.
- Tables give $P(X \leq r)$; deduce single values, tails and intervals by subtraction, watching the $r - 1$ shifts.
- $P(X \geq 1) = 1 - e^{-\lambda}$.
- Independent sums: $\text{Po}(\lambda) + \text{Po}(\mu) = \text{Po}(\lambda + \mu)$; the difference is never Poisson.

6 POISSON APPROXIMATION TO THE BINOMIAL AND LINK WITH FREQUENCY DISTRIBUTIONS

In the previous sections we discovered the Poisson distribution as a model for rare events occurring at random in time or space: calls to a call centre, accidents at a junction on the Yaoundé–Douala road, misprints per page. We also noticed something striking: the Poisson probabilities $P(X = r) = e^{-\lambda}\lambda^r / r!$ are far easier to compute than binomial probabilities, which involve large binomial coefficients $\binom{n}{r}$. This final section exploits that observation. We shall see that when n is large and p is small, the binomial distribution $B(n, p)$ is almost indistinguishable from a Poisson distribution with the same mean $\lambda = np$. We then study an important property of the Poisson distribution — the distribution of the sum of two independent Poisson variables — before closing the chapter by connecting all our theoretical distributions to real data through frequency distributions, which is exactly how a statistician checks whether a model fits reality.

6.1 Why We Need an Approximation

Imagine a factory producing $n = 2000$ bottle tops per day, each defective with probability $p = 0.002$. The number X of defective tops follows $B(2000, 0.002)$. To find $P(X = 5)$ exactly we need

$$P(X = 5) = \binom{2000}{5}(0.002)^5(0.998)^{1995}.$$

The binomial coefficient $\binom{2000}{5}$ has more than a dozen digits, and $(0.998)^{1995}$ is an extremely small number requiring careful handling. For a GCE candidate with a basic calculator, and even for a computer in extreme cases, this is heavy and error-prone. Yet the mean of this distribution is simply $np = 4$, a perfectly moderate number. The idea of the Poisson approximation is: when events are numerous but individually rare, only the mean matters, and the Poisson law with $\lambda = np$ reproduces the binomial probabilities to excellent accuracy.

PROPERTY **Poisson approximation to the binomial**

Let $X \sim B(n, p)$. If n is **large** and p is **small** so that the mean $\lambda = np$ is **moderate**, then

$$X \text{ is approximately } \mathrm{Po}(np), \qquad \text{i.e.} \qquad P(X = r) \approx e^{-np}\,\frac{(np)^r}{r!}.$$

Rule of thumb used at GCE: the approximation is acceptable when $n \geq 50$ and $p \leq 0.1$ (some texts use $np \leq 10$ as an additional safeguard). The approximation improves as n grows and p shrinks with np fixed.

WARNING **The variance changes slightly**

The approximation preserves the **mean**: both $B(n, p)$ and $\mathrm{Po}(np)$ have mean np. But the binomial variance is npq while the Poisson variance is np. Since $q = 1 - p$, the two variances agree only when $q \approx 1$, i.e. when p is small. This is precisely why the condition $p \leq 0.1$ mat–

ters: with $p = 0.1$, the variance is underestimated by about 10%; with $p = 0.5$ the approximation would be disastrous.

6.2 Heuristic Justification (Enrichment for Strong Candidates)

EXPERIMENT **From the binomial formula to the Poisson formula**

Fix $\lambda > 0$ and let $X_n \sim \mathrm{B}(n, \lambda/n)$, so that $np = \lambda$ for every n. For a fixed integer r,

$$\begin{aligned} P(X_n = r) &= \binom{n}{r}\left(\frac{\lambda}{n}\right)^r\left(1-\frac{\lambda}{n}\right)^{n-r} \\ &= \frac{n(n-1)\cdots(n-r+1)}{r!}\cdot\frac{\lambda^r}{n^r}\cdot\left(1-\frac{\lambda}{n}\right)^n\left(1-\frac{\lambda}{n}\right)^{-r} \\ &= \frac{\lambda^r}{r!}\cdot\underbrace{\frac{n(n-1)\cdots(n-r+1)}{n^r}}_{\to 1}\cdot\underbrace{\left(1-\frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}}\cdot\underbrace{\left(1-\frac{\lambda}{n}\right)^{-r}}_{\to 1}. \end{aligned}$$

The first bracket is a product of r factors $\left(1-\frac{k}{n}\right)$, each tending to 1; the classical limit $\left(1-\frac{\lambda}{n}\right)^n \to e^{-\lambda}$ gives the exponential; the last bracket tends to 1. Hence

$$P(X_n = r) \longrightarrow e^{-\lambda}\frac{\lambda^r}{r!} \quad \text{as } n \to \infty,$$

which is exactly the Poisson probability. This derivation is not examinable, but it explains why the rule of thumb works: for large n and small p, the binomial formula is already "almost" the Poisson formula.

6.3 Sum of Independent Poisson Variables (Lesson 132)

A fundamental property of the Poisson distribution — often used in conjunction with the approximation — is that the sum of independent Poisson variables is again Poisson. This makes the Poisson family closed under addition, a fact that simplifies many real-world models (e.g., total calls from two independent switchboards).

PROPERTY **Sum of independent Poisson variables**

If $X \sim \mathrm{Po}(\lambda)$ and $Y \sim \mathrm{Po}(\mu)$ are independent, then

$$X + Y \sim \mathrm{Po}(\lambda + \mu).$$

Proof (examinable): Use the moment generating function (MGF) or the convolution formula. MGF: $M_X(t) = e^{\lambda(e^t-1)}$, $M_Y(t) = e^{\mu(e^t-1)}$. By independence, $M_{X+Y}(t) = M_X(t)M_Y(t) = e^{(\lambda+\mu)(e^t-1)}$, which is the MGF of $\mathrm{Po}(\lambda+\mu)$. Convolution: $P(X+Y=r) = \sum_{k=0}^{r} P(X=k)P(Y=r-k) =$

$$\sum_{k=0}^{r} e^{-\lambda}\frac{\lambda^k}{k!}e^{-\mu}\frac{\mu^{r-k}}{(r-k)!} = e^{-(\lambda+\mu)}\frac{1}{r!}\sum_{k=0}^{r}\binom{r}{k}\lambda^k\mu^{r-k} = e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^r}{r!}.$$

EXEMPLE **Combining Poisson processes**

The number of emails received by a teacher follows Po(3) per hour, and the number of SMS messages follows Po(2) per hour, independently. The total number of electronic messages received in one hour follows Po(3 + 2) = Po(5). The probability of receiving exactly 4 messages in an hour is $e^{-5}5^4/4! \approx 0.1755$.

6.4 Worked Comparison: Exact Binomial versus Poisson Approximation

WORKED EXAMPLE 13 **Comparing** B(100, 0.04) **with** Po(4)

A machine produces 100 items per batch; each item is defective independently with probability 0.04. Let X be the number of defective items in a batch. (a) Compute the exact $P(X = 2)$ and $P(X = 5)$. (b) Compute the Poisson approximations. (c) Comment.

SOLUTION

Solution. Here $n = 100 \geq 50$, $p = 0.04 \leq 0.1$ and $np = 4$: the rule of thumb is satisfied.
(a) Exact binomial values.

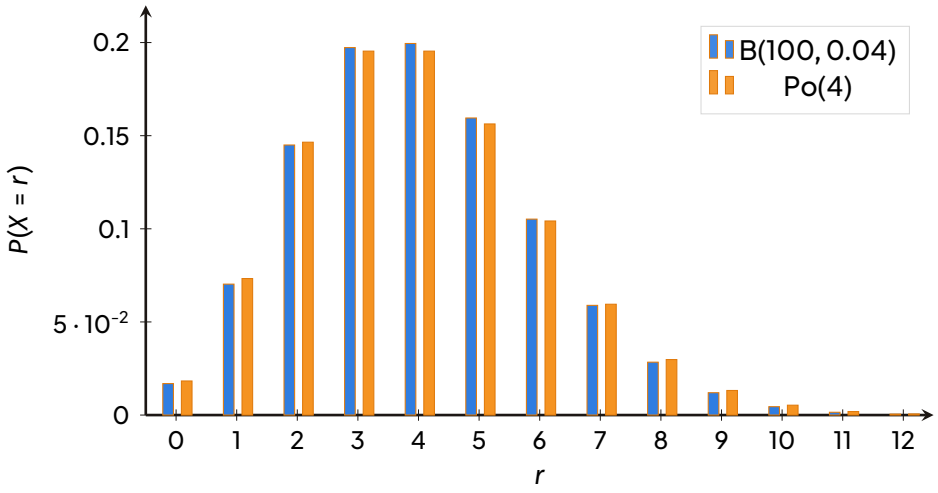
$$P(X = 2) = \binom{100}{2}(0.04)^2(0.96)^{98} = 4950 \times 0.0016 \times (0.96)^{98} \approx 0.1450,$$
$$P(X = 5) = \binom{100}{5}(0.04)^5(0.96)^{95} \approx 0.1595.$$

(b) Poisson approximation with $\lambda = np = 4$.

$$P(X = 2) \approx e^{-4}\frac{4^2}{2!} = 8e^{-4} \approx 0.1465,$$
$$P(X = 5) \approx e^{-4}\frac{4^5}{5!} = \frac{1024}{120}e^{-4} \approx 0.1563.$$

(c) Comment. The absolute errors are about 0.0015 and 0.0032 — roughly 1–2% relative error. For practical purposes (and for GCE marking) the approximation is excellent, and it required no binomial coefficients at all.

The figure below displays the two distributions side by side: the bars are barely distinguishable.



Exact binomial probabilities B(100, 0.04) (blue) and Poisson approximation Po(4) (orange). The two distributions share the same mean 4 and are almost identical.

METHOD **Applying the Poisson approximation in an exam**

1. Check the conditions: $n \geq 50$, $p \leq 0.1$ (and ideally $np \leq 10$).
2. State the approximation explicitly: "$X \sim \mathrm{B}(n, p) \approx \mathrm{Po}(\lambda)$ with $\lambda = np = \ldots$".
3. Compute with the Poisson formula $P(X = r) = e^{-\lambda}\lambda^r / r!$ or the cumulative Poisson tables (for $P(X \leq r)$).
4. Conclude in the context of the question.

WARNING **Exam tip: name your approximation**

If a question says "use a suitable approximation", you must write down the approximating distribution and its parameter before any calculation, e.g. "Since $n = 400$ is large and $p = 0.01$ is small, $X \approx \mathrm{Po}(4)$." Markers award a specific mark for this statement; a bare numerical answer without it loses credit.

6.5 Choosing the Right Model: Binomial, Geometric, Poisson or Approximation

A recurring GCE skill is recognising which distribution a situation describes. Use the following decision criteria.

Situation	Model	Key signal
Fixed number n of independent trials, count of successes, p constant	B(n, p)	"out of n …how many …"
Trials repeated until the first success	Geometric; $P(X = r) = q^{r-1}p$	"first time", "until one succeeds"
Events at random in time/space, known average rate λ per unit interval	Po(λ)	"on average λ per hour/page/km"
Binomial with $n \geq 50$, $p \leq 0.1$	Po(np) approximation	"use a suitable approximation"

EXEMPLE **Quick classification**

- A student answers 20 multiple-choice questions at random, each with 4 options: number correct ~ B(20, 0.25) — not approximable by Poisson since $p = 0.25 > 0.1$.
- A typist makes on average 3 errors per page: errors per page ~ Po(3).
- A player rolls a die until the first six: number of rolls is geometric with $p = \frac{1}{6}$.
- A lottery sells 5000 tickets, each winning the jackpot with probability 0.0002: number of jackpot winners ~ B(5000, 0.0002) ≈ Po(1).

6.6 Probability Distributions and Frequency Distributions (Lesson 133)

So far our distributions have been theoretical: they predict probabilities. Real life produces observed frequencies: a teacher counts how many of her 40 students scored 0, 1, 2, … points on a quiz; a statistician counts how many league matches ended with 0, 1, 2, … goals. The bridge between the two worlds is the following fundamental idea.

DEFINITION **Expected (theoretical) frequency**

If a random variable X has $P(X = r) = p_r$ and an experiment is repeated N times indepen-

dently, the **expected frequency** of the value r is

$$E_r = N \times P(X = r) = N\,p_r.$$

The set of pairs (r, E_r) is the theoretical frequency distribution associated with the probability model.

This is simply the statement "probability = long-run proportion": if $P(X = 2) = 0.20$, then out of $N = 150$ repetitions we expect about $150 \times 0.20 = 30$ occurrences of the value 2.

WARNING **Expected frequencies need not be integers**

E_r is an average, not an actual count. Values like 23.7 or 8.42 are perfectly correct. Do not round them prematurely: keep at least one or two decimal places throughout, otherwise accumulated rounding errors distort any later comparison with observed data.

6.7 Estimating the Parameter from Data

In practice the parameter of the model is unknown and must be estimated from the data. The natural estimator is the **sample mean**:

- for a **Poisson** model, λ is the mean, so we take $\hat{\lambda} = \bar{x}$;
- for a **binomial** model with known n, the mean is np, so we take $\hat{p} = \bar{x}/n$.

We then compute the probabilities $P(X = r)$ with this estimated parameter and multiply by N.

METHOD **Fitting a distribution to observed data**

1. Compute the sample mean $\bar{x} = \dfrac{\sum f_r x_r}{\sum f_r}$ from the observed frequency table.
2. Estimate the parameter: $\hat{\lambda} = \bar{x}$ (Poisson) or $\hat{p} = \bar{x}/n$ (binomial).
3. Compute the probabilities $p_r = P(X = r)$ using the fitted model (formula or tables).
4. Multiply by the total frequency: $E_r = N\,p_r$.
5. Compare observed and expected frequencies in a two-row table and comment on the closeness of the fit.

WORKED EXAMPLE 14 **Fitting a Poisson model to goals per match**

Over a season, $N = 120$ matches of a Cameroonian league championship gave the following numbers of goals per match:

Goals r	0	1	2	3	4	5 or more
Observed frequency O_r	14	31	38	22	11	4

Fit a Poisson distribution and compute the expected frequencies.

SOLUTION

Solution. Step 1 — sample mean.

$$\bar{x} = \frac{0(14) + 1(31) + 2(38) + 3(22) + 4(11) + 5(4)}{120} = \frac{31 + 76 + 66 + 44 + 20}{120} = \frac{237}{120} = 1.975.$$

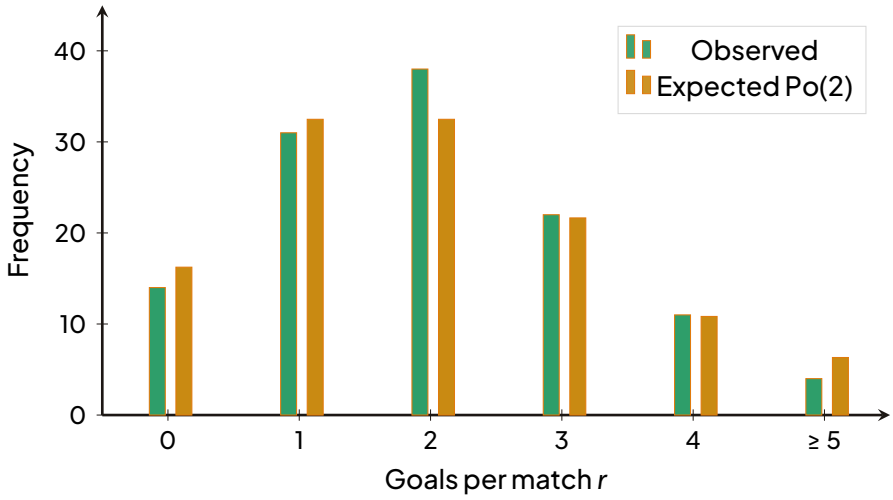
(The "5 or more" class is counted as 5; with such a small frequency the effect is negligible.)
Step 2 — estimate: $\hat{\lambda} = \bar{x} \approx 1.975 \approx 2.0$.
Step 3 — probabilities from Po(2): using $P(X = r) = e^{-2}\,2^r / r!$ with $e^{-2} \approx 0.1353$:

$$p_0 = 0.1353, \quad p_1 = 0.2707, \quad p_2 = 0.2707, \quad p_3 = 0.1804, \quad p_4 = 0.0902,$$

and $P(X \geq 5) = 1 - (p_0 + p_1 + p_2 + p_3 + p_4) \approx 1 - 0.9473 = 0.0527$.
Step 4 — expected frequencies $E_r = 120\,p_r$:

Goals r	0	1	2	3	4	≥ 5
Observed O_r	14	31	38	22	11	4
Expected E_r	16.24	32.48	32.48	21.65	10.83	6.32

Step 5 — comment. The expected frequencies track the observed ones closely (e.g. 38 observed versus 32.5 expected at $r = 2$; 22 versus 21.7 at $r = 3$). A Poisson model with mean about 2 goals per match describes this championship reasonably well. Note that the expected frequencies were deliberately not rounded to integers.



Observed frequencies of goals per match (green) against expected frequencies from the fitted Po(2) model (gold). The two profiles are close, supporting the Poisson model.

DID YOU KNOW? **Towards goodness-of-fit tests**

Comparing observed and expected frequencies "by eye", as we did above, is the first step of a much deeper theory. In further study you will meet the chi-squared goodness-of-fit test, which turns the discrepancies $O_r - E_r$ into a single number $\sum (O_r - E_r)^2 / E_r$ and decides objectively whether the model is acceptable. This is beyond the GCE syllabus, but everything you have done here — estimating λ, computing expected frequencies, keeping decimals — is exactly its preparation.

EXERCISE 6 **Binomial fit, Poisson approximation and Sum of Poissons**

1. In 80 families with 4 children, the number of boys was counted, giving mean 2.05 boys per family. Assuming a binomial model B(4, p), estimate p and compute the expected frequencies of 0, 1, 2, 3, 4 boys among the 80 families.

2. A call centre receives on average 1.5 calls per minute. Calls arrive independently. Use a Poisson model to find the expected number of minutes with no call in one hour (60 minutes).

3. The number of defective components in boxes of 200 is modelled by B(200, 0.015). State a suitable approximation and use it to estimate $P(X \leq 2)$.

4. The number of accidents on Road A follows Po(2) per month, and on Road B follows Po(3) per month, independently. Find the probability that there are exactly 4 accidents in total on both roads in a given month.

SOLUTION **Exercise 1**

1. $\hat{p} = \bar{x}/n = 2.05/4 = 0.5125$. With B(4, 0.5125): $p_0 = (0.4875)^4 \approx 0.0565$, $p_1 = 4(0.5125)(0.4875)^3 \approx 0.2376$, $p_2 = 6(0.5125)^2(0.4875)^2 \approx 0.3745$, $p_3 = 4(0.5125)^3(0.4875) \approx 0.2625$, $p_4 = (0.5125)^4 \approx 0.0690$. Expected frequencies (×80): 4.52, 19.01, 29.96, 21.00, 5.52.
2. Calls per minute ~ Po(1.5), so $P(X = 0) = e^{-1.5} \approx 0.2231$. Expected number of quiet minutes: 60 × 0.2231 ≈ 13.4 minutes.
3. $n = 200 \geq 50$, $p = 0.015 \leq 0.1$, $np = 3$: approximate by Po(3). Then $P(X \leq 2) \approx e^{-3}\left(1 + 3 + \frac{9}{2}\right) = 8.5\,e^{-3} \approx 0.4232$ (confirmed by cumulative Poisson tables).
4. Let $X \sim$ Po(2), $Y \sim$ Po(3) independent. Total $X + Y \sim$ Po(5). $P(X + Y = 4) = e^{-5}5^4/4! = \frac{625}{24}e^{-5} \approx 0.1755$.

KEY POINTS **Key points of the section**

- B$(n, p) \approx$ Po(np) when $n \geq 50$ and $p \leq 0.1$; the mean np is preserved, the variance changes from npq to np.
- If $X \sim$ Po(λ) and $Y \sim$ Po(μ) are independent, then $X + Y \sim$ Po$(\lambda + \mu)$.
- In an exam, state the approximating distribution and its parameter before computing.
- Choose the model from the situation: fixed n trials → binomial; waiting for the first success → geometric; random events at a rate → Poisson; large n, small p → Poisson approximation.
- Expected frequency = $N \times P(X = r)$; estimate the parameter from the sample mean ($\hat{\lambda} = \bar{x}$ or $\hat{p} = \bar{x}/n$).
- Expected frequencies are not rounded; compare them with observed frequencies to judge the fit of the model.

7 INTEGRATION ACTIVITY

8 SPECIAL DISCRETE PROBABILITY DISTRIBUTIONS

8.1 Aims of the Activity

By the end of this activity, you should be able to:

- recognise, from a concrete description, which of the **binomial**, **Poisson**, **geometric** or **uniform** models is appropriate, and justify the choice by checking the conditions of each model;
- estimate the parameter λ of a Poisson distribution from observed data and compare observed frequencies with expected frequencies;
- compute means, variances and probabilities for $B(n, p)$, and use the **Poisson approximation** $Po(np)$ when n is large and p is small;
- use **cumulative tables** (binomial and Poisson) efficiently;
- apply the **unit-interval scaling** of the Poisson mean ($\lambda \propto$ length of interval);
- communicate a complete modelling argument in clear mathematical English, as required at the GCE Advanced Level.

> **WARNING** **Working in Groups**
>
> Work in groups of three or four. Every numerical answer must be accompanied by one sentence stating which model was used and why. An unjustified number earns no credit in this activity, exactly as in the examination.

8.2 Situation

The Douala City Council, together with a local taxi agency "Wouri Express", has commissioned your Further Mathematics class to analyse three sets of data collected during the first term.

Data set A — Accidents at Rond-point Deïdo. A traffic officer recorded the number of road accidents occurring each day at the Rond-point Deïdo junction over 100 consecutive days. Accidents on different days may be assumed independent, and they occur randomly at an approximately constant average rate. The observed frequencies are:

Number of accidents per day x	0	1	2	3	4	5	6	7 or more
Observed frequency (days)	15	31	26	17	8	2	1	0

The sample mean is $\bar{x} = 1.82 \approx 1.8$ accidents per day.

Data set B — Expired insurance at a checkpoint. At a police checkpoint on the Douala–Yaoundé road, long-term records show that 2% of vehicles have expired insurance. Each day, exactly 60 vehicles are checked, independently of one another. Let X be the number of vehicles found with expired insurance in one day, and let T be the number of vehicles checked up to and including the first uninsured vehicle.

Data set C — Calls at Wouri Express. The taxi agency receives booking calls randomly, at a constant average rate of 3 calls per 5 minutes, calls arriving independently of one another. Let Y be the number of calls received in 5 minutes, and W the number received in 2 minutes.

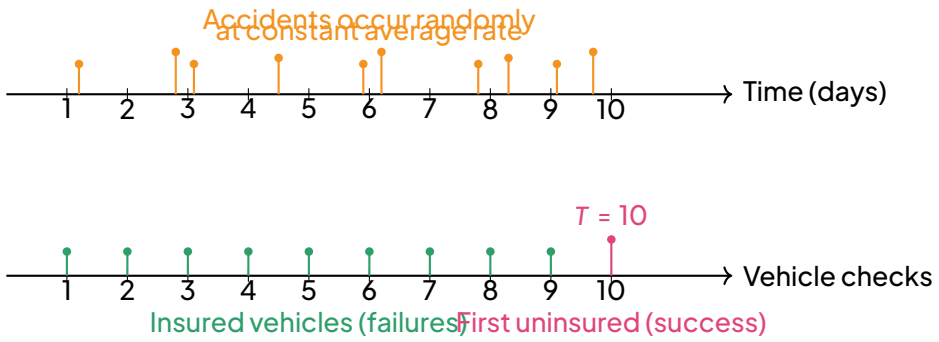
The Council wants a written report answering the tasks below.

8.3 Tasks

1. **Choosing a model for accidents.** State, with reasons, why a Poisson distribution is a plausible model for the number of accidents per day. Estimate the parameter λ from the data.
2. **Expected frequencies.** Using $Po(1.8)$ and the cumulative Poisson tables (or the recurrence relation $P(X = x+1) = \frac{\lambda}{x+1}P(X = x)$), compute $P(X = x)$ for $x = 0, 1, \ldots, 6$ and $P(X \geq 7)$, each to 4 decimal places. Hence compute the expected frequencies out of 100 days and display observed and expected frequencies in one table. Comment on the fit.
3. **Binomial model for uninsured vehicles.** Justify that $X \sim B(60, 0.02)$. Write down $E(X)$ and $\mathrm{Var}(X)$. Compute $P(X = 0)$ and $P(X \geq 2)$ (you may use cumulative binomial tables or direct calculation).
4. **Poisson approximation.** Explain why X may be approximated by $Po(1.2)$. Recompute $P(X = 0)$ and $P(X \geq 2)$ under this approximation and comment on the closeness of the results.
5. **Geometric model.** Explain why T follows a geometric distribution with $p = 0.02$. Find the probability that the first uninsured vehicle appears after the 20th check, i.e. $P(T > 20)$. Write down $E(T)$ and interpret it in context.
6. **Calls and unit intervals.** State the distributions of Y and W, justifying the value of the mean of W. Using cumulative Poisson tables, find $P(Y > 5)$ and $P(W = 0)$.
7. **Synthesis.** Write a short paragraph (5–8 lines) for the City Council summarising your findings: the fit of the accident model, the expected number of uninsured vehicles per day, and the busiest risk the taxi agency should prepare for.

METHOD **Checklist Before Choosing a Model**

1. Fixed number n of independent trials, two outcomes, constant p? ⇒ Binomial.
2. Independent events at a constant average rate in time/space? ⇒ Poisson.
3. Number of trials until the first success? ⇒ Geometric.
4. Binomial with n large ($n \geq 50$) and p small ($p \leq 0.1$)? ⇒ approximate by $Po(np)$.



Left: Poisson process for daily accidents. Right: Geometric setting for first uninsured vehicle.

8.4 Model Answers

WORKED EXAMPLE 15 **Task 1: Model for Accidents**

Statement: State, with reasons, why a Poisson distribution is a plausible model for the number of accidents per day. Estimate the parameter λ from the data.

SOLUTION

Detailed Solution:
The Poisson distribution models the number of events occurring in a fixed interval of time (or space) when the following four conditions hold:

1. Randomness: Events occur at random, not at fixed intervals.
2. Independence: The occurrence of an event in one interval does not affect the probability of an event in another non-overlapping interval. (Accidents on different days are independent.)
3. Constant average rate: The average number of events per unit interval is constant. (The problem states "approximately constant average rate".)
4. Singly: Two events cannot occur at exactly the same instant; in practice, the probability of more than one accident in a very short sub-interval is negligible.

The description of Data Set A explicitly mentions independence and a constant average rate, and accidents are random, discrete events. Therefore, a Poisson distribution is perfectly appropriate.
For a Poisson random variable $X \sim Po(\lambda)$, the expectation is $E(X) = \lambda$. The standard estimator for λ is the sample mean $\bar{x}$.

$$\hat{\lambda} = \bar{x} = \frac{\sum x \cdot f}{\sum f} = \frac{0(15) + 1(31) + 2(26) + 3(17) + 4(8) + 5(2) + 6(1)}{100} = \frac{182}{100} = 1.82.$$

We round to one decimal place as suggested: $\hat{\lambda} \approx 1.8$. Thus we fit the model $X \sim Po(1.8)$.

WORKED EXAMPLE 16 **Task 2: Expected Frequencies and Goodness of Fit**

Statement: Using $Po(1.8)$ and the cumulative Poisson tables (or the recurrence relation), compute $P(X = x)$ for $x = 0, 1, \ldots, 6$ and $P(X \geq 7)$, each to 4 decimal places. Hence compute the expected frequencies out of 100 days and display observed and expected frequencies in one table. Comment on the fit.

SOLUTION

Detailed Solution:

We use the recurrence relation for the Poisson distribution:

$$P(X = 0) = e^{-\lambda} = e^{-1.8} = 0.165298\ldots \approx 0.1653,$$

$$P(X = x + 1) = \frac{\lambda}{x + 1} P(X = x).$$

$$P(X = 1) = \frac{1.8}{1} \times 0.1653 = 0.29754 \approx 0.2975,$$

$$P(X = 2) = \frac{1.8}{2} \times 0.2975 = 0.26775 \approx 0.2678,$$

$$P(X = 3) = \frac{1.8}{3} \times 0.2678 = 0.16068 \approx 0.1607,$$

$$P(X = 4) = \frac{1.8}{4} \times 0.1607 = 0.072315 \approx 0.0723,$$

$$P(X = 5) = \frac{1.8}{5} \times 0.0723 = 0.026028 \approx 0.0260,$$

$$P(X = 6) = \frac{1.8}{6} \times 0.0260 = 0.00780 \approx 0.0078.$$

Sum of probabilities for $x = 0$ to 6:

$$\sum_{x=0}^{6} P(X = x) = 0.1653 + 0.2975 + 0.2678 + 0.1607 + 0.0723 + 0.0260 + 0.0078 = 0.9974.$$

Hence

$$P(X \geq 7) = 1 - 0.9974 = 0.0026.$$

Expected frequencies are obtained by multiplying each probability by 100 (the total number of days).

x	0	1	2	3	4	5	6	≥ 7
Observed frequency	15	31	26	17	8	2	1	0
Expected frequency	16.5	29.8	26.8	16.1	7.2	2.6	0.8	0.3

Comment on the fit: The expected frequencies are very close to the observed frequencies across all categories. The largest discrepancies are at $x = 0$ (observed 15 vs expected 16.5) and $x = 1$ (observed 31 vs expected 29.8), but these differences are small relative to the frequencies. The model $Po(1.8)$ provides an excellent fit to the accident data. (A formal χ^2 goodness-of-fit test would confirm this; it is beyond our syllabus, but the visual agreement is already convincing.)

WORKED EXAMPLE 17 **Task 3: Binomial Model for Uninsured Vehicles**

Statement: Justify that $X \sim B(60, 0.02)$. Write down $E(X)$ and $\mathrm{Var}(X)$. Compute $P(X = 0)$ and $P(X \geq 2)$.

SOLUTION

Detailed Solution:
Justification for Binomial model:

- There is a fixed number of trials: $n = 60$ vehicles checked each day.
- Each trial has two outcomes: "expired insurance" (success) or "valid insurance" (failure).
- The probability of success is constant: $p = 0.02$ (2% of vehicles have expired insurance).
- The trials are independent: the insurance status of one vehicle does not affect another.

Therefore X, the number of vehicles with expired insurance in a day, follows a binomial distribution: $X \sim B(60, 0.02)$.
Mean and Variance:

$$E(X) = np = 60 \times 0.02 = 1.2,$$

$$\mathrm{Var}(X) = np(1 - p) = 60 \times 0.02 \times 0.98 = 1.176.$$

Probabilities:

$$P(X = 0) = (1 - p)^n = 0.98^{60} = 0.29755\ldots \approx 0.2976 \quad \textbf{(4 d.p.).}$$

$$P(X = 1) = \binom{60}{1} p(1 - p)^{59} = 60 \times 0.02 \times 0.98^{59} = 1.2 \times 0.30363\ldots = 0.36436\ldots \approx 0.3644.$$

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.2976 - 0.3644 = 0.3380.$$

(Using cumulative binomial tables with $n = 60, p = 0.02$ gives the same result.)

WORKED EXAMPLE 18 **Task 4: Poisson Approximation to the Binomial**

Statement: Explain why X may be approximated by $Po(1.2)$. Recompute $P(X = 0)$ and $P(X \geq 2)$ under this approximation and comment on the closeness of the results.

SOLUTION

Detailed Solution:
The Poisson approximation to the binomial distribution is valid when:

- n is large (typically $n \geq 50$),

- p is small (typically $p \leq 0.1$),
- np is moderate (typically $np \leq 5$).

Here $n = 60 \geq 50$, $p = 0.02 \leq 0.1$, and $np = 1.2 \leq 5$. All conditions are satisfied. The approximating Poisson distribution has parameter $\lambda = np = 1.2$, so $X \approx Po(1.2)$.
Approximate probabilities:

$$P(X = 0) = e^{-1.2} = 0.30119\ldots \approx 0.3012.$$

$P(X \geq 2) = 1-P(X = 0)-P(X = 1) = 1-e^{-1.2}-1.2e^{-1.2} = 1-e^{-1.2}(1+1.2) = 1-0.3012\times2.2 = 1-0.66264 = 0.33736 \approx 0.3374$

Comparison:

Probability	Exact Binomial	Poisson Approximation	Absolute Error
$P(X = 0)$	0.2976	0.3012	0.0036
$P(X \geq 2)$	0.3380	0.3374	0.0006

The errors are very small (less than 0.004). The Poisson approximation is excellent here and much quicker to use with tables, especially when n is large and binomial tables for $n = 60$ may not be available.

WORKED EXAMPLE 19 **Task 5: Geometric Model for First Uninsured Vehicle**

Statement: Explain why T follows a geometric distribution with $p = 0.02$. Find $P(T > 20)$. Write down $E(T)$ and interpret it in context.

SOLUTION

Detailed Solution:
Justification for Geometric model: The random variable T counts the number of vehicles checked up to and including the first uninsured vehicle. This is exactly the setting of the geometric distribution:

- We perform independent Bernoulli trials (checking vehicles).
- Each trial has the same success probability $p = 0.02$ (expired insurance).
- We stop at the first success.
- T is the total number of trials needed to obtain the first success.

Hence $T \sim Geo(0.02)$, with probability mass function $P(T = t) = (1 - p)^{t-1}p = 0.98^{t-1} \times 0.02$ for $t = 1, 2, 3, \ldots$
Probability that the first uninsured vehicle appears after the 20th check: The event $\{T > 20\}$ means the first 20 vehicles all have valid insurance (i.e., 20 consecutive failures).

$$P(T > 20) = (1 - p)^{20} = 0.98^{20} = 0.667607\ldots \approx 0.6676 \quad (4\text{ d.p.}).$$

Expectation and interpretation:

$$E(T) = \frac{1}{p} = \frac{1}{0.02} = 50.$$

Interpretation: On average, the checkpoint officers will need to inspect 50 vehicles before finding the first one with expired insurance. This is a long-run average; on any given day the actual number may be much smaller or larger.

WORKED EXAMPLE 20 **Task 6: Calls and Unit Interval Scaling**

Statement: State the distributions of Y and W, justifying the value of the mean of W. Using cumulative Poisson tables, find $P(Y > 5)$ and $P(W = 0)$.

SOLUTION

Detailed Solution:
Distribution of Y (calls in 5 minutes): Calls arrive randomly, independently, and at a constant average rate. The number of calls in a fixed time interval follows a Poisson distribution. The rate is 3 calls per 5 minutes, so for a 5-minute interval:

$$Y \sim Po(3).$$

Distribution of W (calls in 2 minutes): The Poisson mean is proportional to the length of the time interval. The rate is 3 calls per 5 minutes, so the rate per minute is $3/5 = 0.6$ calls/minute. For a 2-minute interval:

$$\lambda_W = 0.6 \times 2 = 1.2 \quad \text{or directly} \quad \lambda_W = 3 \times \frac{2}{5} = 1.2.$$

Thus $W \sim Po(1.2)$.
Probabilities from cumulative Poisson tables:

- For $Y \sim Po(3)$: $P(Y > 5) = 1 - P(Y \leq 5)$. From tables, $P(Y \leq 5) = 0.9161$. Hence $P(Y > 5) = 1 - 0.9161 = 0.0839$.
- For $W \sim Po(1.2)$: $P(W = 0) = e^{-1.2} = 0.3012$ (directly from formula or tables).

WORKED EXAMPLE 21 **Task 7: Synthesis for the City Council**

Statement: Write a short paragraph (5–8 lines) summarising your findings.

SOLUTION

Detailed Solution:
"The daily number of accidents at Rond-point Deïdo is well modelled by a Poisson distri-

bution with mean 1.8: the expected frequencies match the observed ones closely, so the junction averages just under two accidents per day, with about 17 accident-free days per 100. At the checkpoint, an average of 1.2 vehicles per day (out of 60) have expired insurance, and there is roughly a one-in-three chance (33.8%) of catching at least two offenders in a day; the first offender typically appears around the 50th vehicle checked. Wouri Express should expect busy spells: there is about an 8.4% chance of receiving more than 5 calls in 5 minutes, while a completely silent 2-minute spell occurs with probability about 0.30."

8.5 Going Further

EXERCISE 7 **Extension Questions**

1. Using the fitted model $Po(1.8)$, find the probability of at least one day with 4 or more accidents in a week of 7 independent days. Hint: first find $p = P(X \geq 4)$, then use a binomial model with $n = 7$.
2. The taxi agency merges its calls with those of a partner agency receiving calls independently at rate 2 per 5 minutes. State the distribution of the total number of calls per 5 minutes and compute the probability of exactly 4 calls.

SOLUTION **Extension Questions**

1. From the Poisson tables (or recurrence) with $\lambda = 1.8$:

$P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) = 0.1653 + 0.2975 + 0.2678 + 0.1607 = 0.8913.$

Thus $p = P(X \geq 4) = 1 - 0.8913 = 0.1087$. Let N be the number of days (out of 7) with ≥ 4 accidents. Since days are independent, $N \sim B(7, 0.1087)$.

$$P(N \geq 1) = 1 - P(N = 0) = 1 - (1 - 0.1087)^7 = 1 - 0.8913^7 = 1 - 0.4468 = 0.5532.$$

2. The sum of two independent Poisson random variables is Poisson with mean equal to the sum of the means. Total calls per 5 minutes $\sim Po(3 + 2) = Po(5)$.

$$P(\text{Total} = 4) = \frac{e^{-5} \cdot 5^4}{4!} = \frac{0.0067379 \times 625}{24} = \frac{4.2112}{24} = 0.1755 \quad \text{(4 d.p.)}.$$

KEY POINTS **Key Points Consolidated by This Activity**

- A Poisson parameter λ is estimated by the sample mean; a model is validated by comparing observed and expected frequencies.
- Binomial $B(n, p)$: $E(X) = np$, $\mathrm{Var}(X) = np(1-p)$; approximated by $Po(np)$ for large n ($n \geq 50$), small p ($p \leq 0.1$).
- Geometric $\mathrm{Geo}(p)$: $P(T > k) = (1-p)^k$, $E(T) = \frac{1}{p}$.
- Poisson means scale linearly with the interval length: rate 3 per 5 min $\Rightarrow$ mean 1.2 per 2 min.
- The sum of independent Poisson variables is Poisson with the sum of the means.

DID YOU KNOW? **Did You Know?**

The Poisson distribution is named after Siméon Denis Poisson (1781–1840), a French mathematician who introduced it in 1837. In Cameroon, Poisson processes are used to model:

- Call arrivals at MTN/Orange call centres (teletraffic engineering).
- Customer arrivals at banks (e.g., BICEC, Afriland First Bank) for queue management.
- Defects in manufactured goods (e.g., at SOSUCAM or CICAM factories).
- Rainfall events in Douala during the rainy season for hydrological planning.

The geometric distribution is the discrete analogue of the exponential distribution, which models waiting times between Poisson events.

9 METHODS AND WORKED EXAMPLES

10 SPECIAL DISCRETE PROBABILITY DISTRIBUTIONS

10.1 The Binomial Distribution: Probabilities, Mean and Variance

DEFINITION **Binomial Experiment**

A **binomial experiment** satisfies four conditions:

1. There is a fixed number n of trials.
2. Each trial has exactly two outcomes: success or failure.
3. The probability of success p is constant from trial to trial.
4. The trials are independent.

If X is the number of successes in n trials, we write $X \sim B(n, p)$ and set $q = 1 - p$.

PROPERTY **Probability Mass Function, Mean and Variance of $B(n, p)$**

For $X \sim B(n, p)$ with $q = 1 - p$:

$$P(X = r) = \binom{n}{r} p^r q^{n-r}, \qquad r = 0, 1, \ldots, n,$$

$$E(X) = np,$$

$$\mathrm{Var}(X) = npq,$$

$$\sigma_X = \sqrt{npq}.$$

Derivation of $E(X)$ and $\mathrm{Var}(X)$ (examinable): Write $X = \sum_{i=1}^{n} X_i$ where $X_i \sim$ Bernoulli(p) are independent. Then $E(X_i) = p$, $\mathrm{Var}(X_i) = pq$. By linearity of expectation and independence: $E(X) = \sum E(X_i) = np$, $\mathrm{Var}(X) = \sum \mathrm{Var}(X_i) = npq$.

METHOD **Computing binomial probabilities, mean and variance**

1. Check the four conditions for a binomial model (fixed n, two outcomes, constant p, independence).

2. Identify n and p, write $X \sim B(n, p)$ and $q = 1 - p$.
3. Exact probability: $P(X = r) = \binom{n}{r} p^r q^{n-r}$.
4. Cumulative probabilities: use complement when shorter: $P(X \geq 1) = 1 - P(X = 0) = 1 - q^n$, $P(X \geq r) = 1 - P(X \leq r - 1)$.
5. Quote mean and variance directly: $E(X) = np$, $\mathrm{Var}(X) = npq$, $\sigma = \sqrt{npq}$.

Reflex: "at least one" ⇒ compute $1 - q^n$ first.

WORKED EXAMPLE 22 **Quality control in a Douala factory**

A machine in a Douala bottling factory produces caps of which 5% are defective. A quality controller takes a random sample of 20 caps. Let X be the number of defective caps in the sample.

1. State the distribution of X, justifying your answer.
2. Find the probability that exactly 2 caps are defective.
3. Find the probability that at least one cap is defective.
4. Calculate the mean and the standard deviation of X.

SOLUTION

Solution.

1. There is a fixed number of trials ($n = 20$), each cap is either defective or not, the probability of a defective cap is constant ($p = 0.05$), and the caps are independent (random sample from a large production). Hence
$$X \sim B(20,\ 0.05), \qquad q = 0.95.$$
2. Using the formula with $r = 2$:
$$P(X = 2) = \binom{20}{2}(0.05)^2(0.95)^{18} = 190 \times 0.0025 \times 0.3972 \approx 0.1887.$$
3. Using the complement:
$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.95)^{20} = 1 - 0.3585 = 0.6415.$$
4. Mean and variance:
$$E(X) = np = 20 \times 0.05 = 1, \qquad \mathrm{Var}(X) = npq = 20 \times 0.05 \times 0.95 = 0.95,$$
$$\sigma = \sqrt{0.95} \approx 0.975.$$

So on average one defective cap is found per sample of 20, with standard deviation about 0.97.

WARNING **Common pitfalls with the binomial model**

- Do not use the binomial distribution if sampling is without replacement from a small population (use hypergeometric instead). The binomial is valid when the population is large relative to the sample (rule of thumb: sample size ≤ 10% of population).
- The probability p must be the same for every trial. If p changes, it is not binomial.
- "At least r" means $r, r+1, \dots, n$; "at most r" means $0, 1, \dots, r$.

10.2 Cumulative Binomial Tables and the Mode

METHOD **Reading cumulative binomial tables**

Cumulative tables give $P(X \le r)$ for $X \sim B(n, p)$ for selected values of n, p and r.

1. Locate the block for your n, then the column for your p, then the row for r: the entry is $P(X \le r)$.
2. Single value: $P(X = r) = P(X \le r) - P(X \le r-1)$ (with $P(X \le -1) = 0$).
3. Strict inequalities: $P(X < r) = P(X \le r-1)$.
4. "Greater than": $P(X > r) = 1 - P(X \le r)$ and $P(X \ge r) = 1 - P(X \le r-1)$.
5. If $p > 0.5$, use the **symmetry trick**: if $X \sim B(n, p)$ then $Y = n - X \sim B(n, 1-p)$, so

$$P(X \le r) = P(Y \ge n - r) = 1 - P(Y \le n - r - 1).$$

Reflex: always rewrite the required probability in terms of $P(X \le \cdot)$ before opening the table.

WORKED EXAMPLE 23 **Using the tables for $B(10, 0.25)$**

A candidate answers a multiple-choice quiz of 10 questions by guessing; each question has 4 options, so the probability of a correct answer is 0.25. The number of correct answers is $X \sim B(10, 0.25)$. Extract from the cumulative tables:

r	$P(X \le r)$, $p = 0.25$	r	$P(X \le r)$, $p = 0.25$
0	0.0563	4	0.9219
1	0.2440	5	0.9803
2	0.5256	6	0.9965
3	0.7759	7	0.9996

Find (a) $P(X = 3)$; (b) $P(X < 2)$; (c) $P(X \geq 4)$; (d) $P(2 \leq X \leq 5)$.

SOLUTION

Solution. All probabilities are rewritten using cumulative values.

(a) $P(X = 3) = P(X \leq 3) - P(X \leq 2) = 0.7759 - 0.5256 = 0.2503$.

(b) $P(X < 2) = P(X \leq 1) = 0.2440$.

(c) $P(X \geq 4) = 1 - P(X \leq 3) = 1 - 0.7759 = 0.2241$.

(d) $P(2 \leq X \leq 5) = P(X \leq 5) - P(X \leq 1) = 0.9803 - 0.2440 = 0.7363$.

WARNING

A very common error is to write $P(X \geq 4) = 1 - P(X \leq 4)$. Since X is discrete, $P(X \geq 4) = 1 - P(X \leq 3)$: the boundary value must be excluded carefully.

PROPERTY **Recurrence Formula and Mode of $B(n, p)$**

For $X \sim B(n, p)$ with $q = 1 - p$:

$$\frac{P(X = r + 1)}{P(X = r)} = \frac{n - r}{r + 1} \cdot \frac{p}{q}, \qquad r = 0, 1, \ldots, n - 1.$$

The probabilities increase while $\frac{n - r}{r + 1} \cdot \frac{p}{q} > 1$, i.e. while $r + 1 < (n + 1)p$. Hence the **mode** (most likely value) is

$$m = \lfloor (n + 1)p \rfloor.$$

If $(n + 1)p$ is an integer, there are two modes: $m = (n + 1)p - 1$ and $m = (n + 1)p$.

METHOD **Finding the mode via the recurrence relation**

1. Write down the recurrence formula: $\frac{P(X = r + 1)}{P(X = r)} = \frac{n - r}{r + 1} \cdot \frac{p}{q}$.
2. The probabilities increase while $\frac{n - r}{r + 1} \cdot \frac{p}{q} > 1$, i.e. while $r < (n + 1)p - 1$.
3. Equivalently, compute $(n + 1)p$:
   - If $(n + 1)p$ is not an integer, the mode is $\lfloor (n + 1)p \rfloor$.
   - If $(n + 1)p$ is an integer, there are two modes: $(n + 1)p - 1$ and $(n + 1)p$.

Reflex: compute $(n+1)p$ first; if it is not an integer, take the integer part.

WORKED EXAMPLE 24 **Most likely number of sixes**

A fair die is thrown 12 times. Let X be the number of sixes obtained.

1. Show that $\dfrac{P(X=r+1)}{P(X=r)} = \dfrac{12-r}{5(r+1)}$.
2. Hence find the mode of X.
3. Verify your answer by computing $P(X=1)$, $P(X=2)$ and $P(X=3)$.

SOLUTION

Solution.

1. Here $X \sim B(12, \frac{1}{6})$, so $p=\frac{1}{6}$, $q=\frac{5}{6}$ and $\dfrac{p}{q}=\dfrac{1}{5}$. The recurrence formula gives

$$\frac{P(X=r+1)}{P(X=r)} = \frac{n-r}{r+1}\cdot\frac{p}{q} = \frac{12-r}{r+1}\cdot\frac{1}{5} = \frac{12-r}{5(r+1)}.$$

2. The probabilities increase while $\dfrac{12-r}{5(r+1)} > 1$, i.e. $12-r > 5r+5$, i.e. $7 > 6r$, i.e. $r < \frac{7}{6}$. So the probabilities increase for $r = 0, 1$ (up to $r+1=2$) and decrease afterwards. Alternatively, $(n+1)p = 13\times\frac{1}{6} = \frac{13}{6} \approx 2.17$, so the mode is $m = \lfloor 2.17 \rfloor = 2$.

3. Check:

$$P(X=1) = \binom{12}{1}\left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^{11} = 12\times\tfrac{1}{6}\times 0.1346 \approx 0.2692,$$

$$P(X=2) = \binom{12}{2}\left(\tfrac{1}{6}\right)^{2}\left(\tfrac{5}{6}\right)^{10} = 66\times\tfrac{1}{36}\times 0.1615 \approx 0.2961,$$

$$P(X=3) = \binom{12}{3}\left(\tfrac{1}{6}\right)^{3}\left(\tfrac{5}{6}\right)^{9} = 220\times\tfrac{1}{216}\times 0.1938 \approx 0.1974.$$

Indeed $P(X=2) > P(X=1)$ and $P(X=2) > P(X=3)$, confirming that the mode is $\boxed{2}$: the most likely outcome is exactly two sixes.

KEY POINTS **Key facts for the Binomial Distribution**

- Conditions: fixed n, two outcomes, constant p, independence.
- $X \sim B(n,p) \Rightarrow P(X=r) = \binom{n}{r}p^r q^{n-r}$.

- $E(X) = np$, $\mathrm{Var}(X) = npq$.
- Cumulative tables give $P(X \le r)$; convert all questions to this form.
- Mode $m = \lfloor (n+1)p \rfloor$; two modes if $(n+1)p \in \mathbb{N}$.
- Symmetry: if $p > 0.5$, use $Y = n - X \sim B(n, 1-p)$.

10.3 The Discrete Uniform Distribution

DEFINITION **Discrete Uniform Distribution**

A discrete random variable X follows a **uniform distribution** on the set $\{x_1, x_2, \dots, x_n\}$ if all outcomes are equally likely:

$$P(X = x_i) = \frac{1}{n}, \qquad i = 1, \dots, n.$$

The standard case is X taking values $1, 2, \dots, n$.

PROPERTY **Mean and Variance of the Uniform Distribution on $\{1, 2, \dots, n\}$**

$$E(X) = \frac{n+1}{2}, \qquad \mathrm{Var}(X) = \frac{n^2-1}{12}.$$

Derivation (examinable):

$$E(X) = \frac{1}{n}\sum_{r=1}^{n} r = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2},$$

$$E(X^2) = \frac{1}{n}\sum_{r=1}^{n} r^2 = \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6},$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

$$= \frac{n+1}{12}\left[2(2n+1) - 3(n+1)\right] = \frac{n+1}{12}(n-1) = \frac{n^2-1}{12}.$$

For a uniform distribution on $\{a, a+1, \dots, b\}$ with $n = b - a + 1$ values:

$$E(X) = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{n^2-1}{12} \quad \text{(variance unchanged by translation).}$$

METHOD **Using the uniform distribution on $\{1, 2, \dots, n\}$**

1. Recognise the model: n equally likely outcomes, $P(X = r) = \dfrac{1}{n}$ for $r = 1, \dots, n$.
2. Quote the key results (derivation examinable):

$$E(X) = \frac{n+1}{2}, \qquad \operatorname{Var}(X) = \frac{n^2 - 1}{12}.$$

3. For outcomes $a, a+1, \dots, b$ (with $n = b - a + 1$ values), shift: $E(X) = \dfrac{a+b}{2}$ and the variance is unchanged, $\operatorname{Var}(X) = \dfrac{n^2-1}{12}$.

Key identities for the derivation: $\sum_{r=1}^{n} r = \dfrac{n(n+1)}{2}$ and $\sum_{r=1}^{n} r^2 = \dfrac{n(n+1)(2n+1)}{6}$.

WORKED EXAMPLE 25 **Deriving the variance and applying it**

1. Prove that if X is uniformly distributed on $\{1, 2, \dots, n\}$, then $\operatorname{Var}(X) = \dfrac{n^2-1}{12}$.
2. A fair tetrahedral die has faces numbered $1, 2, 3, 4$. Find the mean and variance of the score.

SOLUTION

Solution.

1. First, $E(X) = \dfrac{1}{n}\sum_{r=1}^{n} r = \dfrac{1}{n} \cdot \dfrac{n(n+1)}{2} = \dfrac{n+1}{2}$. Next,

$$E(X^2) = \frac{1}{n}\sum_{r=1}^{n} r^2 = \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}.$$

Then

$$\begin{aligned}\operatorname{Var}(X) = E(X^2) - [E(X)]^2 &= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} \\ &= \frac{n+1}{12}\big[2(2n+1) - 3(n+1)\big] = \frac{n+1}{12}(4n + 2 - 3n - 3) \\ &= \frac{(n+1)(n-1)}{12} = \frac{n^2-1}{12}.\end{aligned}$$

2. Here $n = 4$: $E(X) = \dfrac{4+1}{2} = 2.5$ and $\operatorname{Var}(X) = \dfrac{16-1}{12} = \dfrac{15}{12} = 1.25$.

DID YOU KNOW? **Historical note**

The discrete uniform distribution is the simplest model of "fairness". It underlies the classical definition of probability (Laplace, 1812): if all outcomes are equally likely, the probability of an event is the ratio of favourable cases to total cases. In Cameroon, the traditional game of Nsolo (or Awale) uses seeds and pits; the initial distribution of seeds can be modelled using uniform assumptions when players choose pits at random.

10.4 The Geometric Distribution

DEFINITION **Geometric Distribution**

The **geometric distribution** models the number of independent Bernoulli trials (success probability p, failure probability $q = 1 - p$) needed to obtain the first success. We write $X \sim \text{Geo}(p)$.

$$P(X = r) = q^{r-1}p, \qquad r = 1, 2, 3, \ldots$$

Note: Some textbooks define the geometric distribution as the number of failures before the first success (support $0, 1, 2, \ldots$). The GCE syllabus uses the "number of trials until first success" definition (support $1, 2, 3, \ldots$). Always check which definition is used.

PROPERTY **Mean, Variance and Tail Probabilities of** $\text{Geo}(p)$

For $X \sim \text{Geo}(p)$ with $q = 1 - p$:

$$E(X) = \frac{1}{p},$$

$$\text{Var}(X) = \frac{q}{p^2},$$

$$P(X > r) = q^r \quad \text{(first } r \text{ trials all fail)},$$

$$P(X \leq r) = 1 - q^r.$$

The mode is always 1 (the distribution decreases geometrically). Derivation of $E(X)$ (examinable idea): $E(X) = \sum_{r=1}^{\infty} rq^{r-1}p = p\frac{d}{dq}\sum_{r=1}^{\infty} q^r = p\frac{d}{dq}\left(\frac{q}{1-q}\right) = \frac{p}{(1-q)^2} = \frac{1}{p}$.

METHOD **Modelling "waiting time until the first success"**

1. Use the geometric distribution when independent trials with constant success probability p are repeated until the first success; X = number of trials needed.

2. Probabilities: $P(X = r) = q^{r-1}p$, $r = 1, 2, 3, \ldots$
3. Tail probabilities (very useful): $P(X > r) = q^r$ (the first r trials all fail).
4. Mean and variance:
$$E(X) = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{q}{p^2}.$$
5. The mode is always 1: the distribution decreases geometrically.

Reflex: check whether the question counts trials up to and including the first success (geometric) or successes in a fixed number of trials (binomial).

WORKED EXAMPLE 26 **First goal scored**

A striker scores with probability 0.2 on each penalty kick, independently of previous kicks. Let X be the number of kicks taken until he scores his first goal.

1. Write down the distribution of X and compute $P(X = 4)$.
2. Find the probability that he needs more than 3 kicks.
3. Find $E(X)$ and $\mathrm{Var}(X)$, and interpret the mean.
4. Find the smallest n such that the probability that he scores within n kicks exceeds 0.9.

SOLUTION

Solution.

1. $X \sim \mathrm{Geo}(0.2)$ with $p = 0.2$, $q = 0.8$.
$$P(X = 4) = q^3 p = (0.8)^3(0.2) = 0.512 \times 0.2 = 0.1024.$$
2. $P(X > 3) = q^3 = (0.8)^3 = 0.512$.
3. $E(X) = \frac{1}{p} = \frac{1}{0.2} = 5$ kicks on average; $\mathrm{Var}(X) = \frac{q}{p^2} = \frac{0.8}{0.04} = 20$. Interpretation: over many sequences of kicks, the average number of kicks needed to score the first goal is 5.
4. $P(X \le n) = 1 - P(X > n) = 1 - (0.8)^n > 0.9 \iff (0.8)^n < 0.1$. Taking logarithms: $n \ln 0.8 < \ln 0.1$, and since $\ln 0.8 < 0$,
$$n > \frac{\ln 0.1}{\ln 0.8} = \frac{-2.3026}{-0.2231} \approx 10.32.$$
The smallest integer is $n = 11$: he needs at most 11 kicks to score with probability greater than 0.9.

WARNING **Geometric vs. Binomial**

- Binomial: fixed number of trials n, count successes. $X \sim B(n, p)$.
- Geometric: no fixed n, repeat until first success, count trials. $X \sim \text{Geo}(p)$.
- If a question says "find the probability that the first success occurs on the 5th trial", it is geometric. If it says "in 10 trials, find the probability of exactly 3 successes", it is binomial.

10.5 The Poisson Distribution and Its Tables

DEFINITION **Poisson Distribution**

The **Poisson distribution** models the number of events occurring randomly, independently, at a constant average rate in a fixed interval of time or space. We write $X \sim \text{Po}(\lambda)$ where $\lambda > 0$ is the mean number of events in the interval.

$$P(X = r) = e^{-\lambda}\frac{\lambda^r}{r!}, \qquad r = 0, 1, 2, \ldots$$

PROPERTY **Mean, Variance and Additive Property of** $\text{Po}(\lambda)$

For $X \sim \text{Po}(\lambda)$:

$$E(X) = \text{Var}(X) = \lambda.$$

Derivation of $E(X)$ (examinable): $E(X) = \sum_{r=0}^{\infty} re^{-\lambda}\frac{\lambda^r}{r!} = e^{-\lambda}\lambda\sum_{r=1}^{\infty}\frac{\lambda^{r-1}}{(r-1)!} = \lambda e^{-\lambda}e^{\lambda} = \lambda$. Additive property: If $X \sim \text{Po}(\lambda_1)$ and $Y \sim \text{Po}(\lambda_2)$ are independent, then

$$X + Y \sim \text{Po}(\lambda_1 + \lambda_2).$$

Proof (using MGF or convolution): $P(X + Y = n) = \sum_{k=0}^{n} P(X = k)P(Y = n - k) = e^{-(\lambda_1+\lambda_2)}\frac{(\lambda_1+\lambda_2)^n}{n!}$.

METHOD **Modelling random events in a fixed interval**

1. Use the Poisson distribution for events occurring randomly, independently, at a constant average rate in a fixed interval: $X \sim \text{Po}(\lambda)$,

$$P(X = r) = e^{-\lambda}\frac{\lambda^r}{r!}, \qquad E(X) = \text{Var}(X) = \lambda.$$

2. Unit interval reflex: if the rate is given per unit but the question concerns a different

interval, rescale λ proportionally (rate λ per minute $\Rightarrow$ mean 5λ in 5 minutes).

3. Cumulative tables give $P(X \le r)$; manipulate exactly as for binomial tables: $P(X = r) = P(X \le r) - P(X \le r - 1)$, $P(X \ge r) = 1 - P(X \le r - 1)$.
4. If $X \sim \text{Po}(\lambda_1)$ and $Y \sim \text{Po}(\lambda_2)$ are independent, then $X + Y \sim \text{Po}(\lambda_1 + \lambda_2)$.

WORKED EXAMPLE 27 **Calls at a call centre in Yaoundé**

A customer-service centre in Yaoundé receives calls randomly at an average rate of 3 calls per 10 minutes.

1. Find the probability of exactly 2 calls in a 10-minute period.
2. Using the cumulative table extract below, find the probability of at least 4 calls in a 10-minute period.
3. Find the probability of exactly 5 calls in a 20-minute period.
4. Independently, the centre receives emails at an average rate of 1.5 per 10 minutes. Find the probability that the total number of contacts (calls plus emails) in 10 minutes is exactly 3.

Table extract for $\lambda = 3$: $P(X \le 2) = 0.4232$, $P(X \le 3) = 0.6472$.

SOLUTION

Solution.

1. With $\lambda = 3$: $P(X = 2) = e^{-3}\dfrac{3^2}{2!} = e^{-3} \times 4.5 \approx 0.04979 \times 4.5 \approx 0.2240$.
2. $P(X \ge 4) = 1 - P(X \le 3) = 1 - 0.6472 = 0.3528$.
3. New interval, new mean: in 20 minutes, $\lambda = 2 \times 3 = 6$.

$$P(X = 5) = e^{-6}\frac{6^5}{5!} = e^{-6} \times \frac{7776}{120} = e^{-6} \times 64.8 \approx 0.002479 \times 64.8 \approx 0.1606.$$

4. Calls $X \sim \text{Po}(3)$, emails $Y \sim \text{Po}(1.5)$, independent, so $T = X + Y \sim \text{Po}(4.5)$.

$$P(T = 3) = e^{-4.5}\frac{4.5^3}{3!} = e^{-4.5} \times \frac{91.125}{6} = e^{-4.5} \times 15.1875 \approx 0.01111 \times 15.1875 \approx 0.1687.$$

DID YOU KNOW? **Siméon Denis Poisson (1781–1840)**

Poisson was a French mathematician who introduced this distribution in 1837 in his work Recherches sur la probabilité des jugements en matière criminelle et en matière civile. He

derived it as a limit of the binomial distribution when $n \to \infty$, $p \to 0$ with $np = \lambda$ constant. The distribution is now ubiquitous in queueing theory, telecommunications (e.g., modelling call arrivals at MTN or Orange call centres in Cameroon), and reliability engineering.

10.6 Poisson Approximation to the Binomial

PROPERTY **Poisson Approximation to $B(n, p)$**

If n is large (typically $n \geq 50$), p is small (typically $p \leq 0.1$), and $np = \lambda$ is moderate (say $\lambda \leq 5$), then

$$B(n, p) \approx \text{Po}(\lambda) \quad \text{with } \lambda = np.$$

Justification (examinable idea): For fixed $\lambda = np$, as $n \to \infty$,

$$\binom{n}{r} p^r q^{n-r} = \frac{n(n-1)\cdots(n-r+1)}{r!} \left(\frac{\lambda}{n}\right)^r \left(1 - \frac{\lambda}{n}\right)^{n-r} \longrightarrow e^{-\lambda} \frac{\lambda^r}{r!}.$$

The approximation works because both distributions then have mean λ and variance $\approx \lambda$ (since $npq = \lambda(1 - p) \approx \lambda$).

METHOD **Approximating $B(n, p)$ by $\text{Po}(np)$**

1. Check the conditions: n large (typically $n \geq 50$) and p small (typically $p \leq 0.1$), so that np is moderate.
2. Set $\lambda = np$ and replace $X \sim B(n, p)$ by $X \approx \text{Po}(\lambda)$.
3. Compute with the Poisson formula or tables; state the approximation clearly in your answer.
4. Justification (examinable idea): for fixed $\lambda = np$, as $n \to \infty$,

$$\binom{n}{r} p^r q^{n-r} \longrightarrow e^{-\lambda} \frac{\lambda^r}{r!}.$$

Reflex: the approximation works because both distributions then have mean λ and variance $\approx \lambda$ (since $npq = \lambda(1 - p) \approx \lambda$).

WORKED EXAMPLE 28 **Rare blood group**

In a certain region, 2% of people have a particular blood group. A mobile clinic screens

150 people. Let X be the number with this blood group.

1. Explain why X may be approximated by a Poisson distribution and state its parameter.
2. Use the approximation to find $P(X = 4)$ and $P(X \leq 2)$.
3. Compare $P(X = 4)$ with the exact binomial value.

SOLUTION

Solution.

1. $n = 150$ is large and $p = 0.02$ is small, with $np = 3$ moderate, so the Poisson approximation is appropriate: $X \approx \text{Po}(3)$.
2. $$P(X = 4) \approx e^{-3}\frac{3^4}{4!} = e^{-3} \times \frac{81}{24} = e^{-3} \times 3.375 \approx 0.04979 \times 3.375 \approx 0.1680.$$
   For the cumulative probability:
   $$P(X \leq 2) \approx e^{-3}\left(1 + 3 + \frac{9}{2}\right) = e^{-3} \times 8.5 \approx 0.04979 \times 8.5 \approx 0.4232.$$
3. The exact binomial value is $\binom{150}{4}(0.02)^4(0.98)^{146} \approx 0.1693$. The Poisson estimate 0.1680 differs by only about 0.0013, a relative error under 1%: the approximation is excellent.

WARNING

Never use the Poisson approximation when p is large (e.g. $p = 0.5$): the variance npq then differs greatly from np and the approximation fails. For large p, apply the trick to the failures (which have small probability q) instead.

10.7 Linking Probability Distributions to Frequency Distributions

METHOD **Fitting a theoretical distribution to observed data**

1. From the data, estimate the parameter: the sample mean $\bar{x}$ estimates np (binomial) or λ (Poisson).
2. Compute the theoretical probabilities $P(X = r)$ for each value r.
3. Multiply by the total frequency N to obtain the **expected frequencies** $E_r = N \times P(X = r)$.
4. Compare observed and expected frequencies (e.g. by inspection or a goodness-of-fit idea) and conclude whether the model is suitable.

WORKED EXAMPLE 29 **Fitting a Poisson distribution to accident data**

The number of accidents per week at a junction in Bamenda was recorded over 100 weeks:

Accidents per week	0	1	2	3	4 or more
Number of weeks	25	38	24	10	3

The mean number of accidents per week is $\bar{x} = 1.3$.

1. Explain why a Poisson model with $\lambda = 1.3$ might be suitable.
2. Calculate the expected frequencies for 0, 1 and 2 accidents per week (to 1 d.p.).
3. Comment on the fit.

SOLUTION

Solution.

1. Accidents occur randomly, independently and (presumably) at a roughly constant average rate over the weeks; the Poisson distribution is the natural model, with λ estimated by the sample mean 1.3.
2. With $e^{-1.3} \approx 0.2725$:

$$E_0 = 100 \times e^{-1.3} = 100 \times 0.2725 = 27.3,$$
$$E_1 = 100 \times e^{-1.3} \times 1.3 = 27.3 \times 1.3 = 35.4,$$
$$E_2 = 100 \times e^{-1.3} \times \frac{1.3^2}{2} = 35.4 \times \frac{1.3}{2} = 23.0.$$

3. Comparing: observed 25, 38, 24 against expected 27.3, 35.4, 23.0. The agreement is close (differences of only 1 to 3 weeks out of 100), so the Poisson model with $\lambda = 1.3$ fits the data well. This illustrates the general principle: a probability distribution is the theoretical limit of a frequency distribution as the number of observations grows.

KEY POINTS **Choosing the right discrete distribution**

Situation	Distribution	Key parameter(s)
Fixed n trials, count successes	Binomial $B(n, p)$	n, p
Wait for first success	Geometric Geo(p)	p
Events in fixed interval, constant rate	Poisson Po(λ)	λ (mean rate × interval)
Equally likely outcomes $1, \ldots, n$	Uniform	n

Approximations: $B(n, p) \approx \text{Po}(np)$ if $n \geq 50$, $p \leq 0.1$. Sum of independent Poissons: $\text{Po}(\lambda_1)$ +

$\text{Po}(\lambda_2) = \text{Po}(\lambda_1 + \lambda_2)$.

EXERCISE 8 **Mixed practice**

1. A fair coin is tossed 10 times. Find the probability of obtaining exactly 6 heads. State the mean and variance of the number of heads.
2. In a large batch of components, 3% are defective. A sample of 50 is taken. Use a Poisson approximation to find the probability that the sample contains at most 2 defectives.
3. The number of emails received per hour by a server follows a Poisson distribution with mean 4. Find the probability that in a 30-minute period the server receives exactly 3 emails.
4. A random variable X is uniformly distributed on the integers $-2, -1, 0, 1, 2$. Find $E(X)$ and $\text{Var}(X)$.
5. A basketball player makes 70% of her free throws. She shoots until she misses. Find the probability that she makes exactly 4 shots before missing. (Careful: define the success event.)

SOLUTION **Mixed practice**

1. $X \sim B(10, 0.5)$. $P(X = 6) = \binom{10}{6}0.5^{10} = 210/1024 \approx 0.2051$. $E(X) = 5$, $\text{Var}(X) = 2.5$.
2. $n = 50$, $p = 0.03$, $np = 1.5$. $X \approx \text{Po}(1.5)$. $P(X \le 2) = e^{-1.5}(1 + 1.5 + 1.5^2/2) = e^{-1.5} \times 3.625 \approx 0.2231 \times 3.625 \approx 0.8088$.
3. 30 min $\Rightarrow \lambda = 2$. $P(X = 3) = e^{-2} \times 8/6 = 4e^{-2}/3 \approx 0.1804$.
4. Values: $-2, -1, 0, 1, 2$ ($n = 5$). Shift of standard uniform on $1..5$ by -3. $E(X) = (1 + 2 + 3 + 4 + 5)/5 - 3 = 3 - 3 = 0$. $\text{Var}(X) = (5^2 - 1)/12 = 24/12 = 2$.
5. "Makes exactly 4 shots before missing" $\Rightarrow$ first miss on 5th trial. Define success = miss ($p = 0.3$). $X \sim \text{Geo}(0.3)$. $P(X = 5) = 0.7^4 \times 0.3 = 0.2401 \times 0.3 = 0.07203$.

11 EXERCISES

11.1 I check my knowledge

EXERCISE 9 **MCQ: spotting the right distribution**

For each situation, choose the correct distribution for X from: binomial, discrete uniform, geometric, Poisson. Justify briefly.

1. X = number of heads in 20 tosses of a fair coin.
2. X = number of the ticket drawn at random from tickets numbered 1 to 50.
3. X = number of attempts needed to score a first goal, each attempt succeeding with probability 0.3.
4. X = number of customers arriving at a bank counter in one hour, arrivals occurring randomly at an average rate of 12 per hour.

EXERCISE 10 **True or false?**

State whether each assertion is true or false, justifying your answer.

1. If $X \sim B(n, p)$, then $\mathrm{Var}(X) = np$.
2. If $X \sim \mathrm{Po}(\lambda)$, then $E(X) = \mathrm{Var}(X)$.
3. If X is uniformly distributed on $\{1, 2, \dots, n\}$, then $E(X) = \dfrac{n+1}{2}$.
4. If $X \sim \mathrm{Po}(2)$ and $Y \sim \mathrm{Po}(3)$ are independent, then $X + Y \sim \mathrm{Po}(6)$.

EXERCISE 11 **Key definitions and formulas**

1. State the four conditions (Bernoulli scheme) under which a random variable follows a binomial distribution.
2. Write down the probability mass function of $X \sim B(n, p)$ and of $Y \sim \mathrm{Po}(\lambda)$.
3. State the recurrence relation linking $P(X = k+1)$ and $P(X = k)$ for $X \sim B(n, p)$, and explain how it is used to locate the mode.
4. State the mean and variance of the geometric distribution with parameter p (number of trials up to and including the first success).

EXERCISE 12 **Direct calculations**

1. A fair die is thrown 12 times and X is the number of sixes obtained. Write down the distribution of X and compute $E(X)$ and $\mathrm{Var}(X)$.
2. Compute $P(X = 2)$, giving your answer to 4 decimal places.
3. $Y \sim \mathrm{Po}(4)$. Compute $P(Y = 0)$ and $P(Y = 3)$ to 4 decimal places.
4. T is uniformly distributed on $\{1, 2, \ldots, 10\}$. Find $E(T)$ and $\mathrm{Var}(T)$.

11.2 I practise

EXERCISE 13 **Binomial: parameters and mode from the moments**

Given $X \sim B(n, p)$ with $E(X) = 6$ and $\mathrm{Var}(X) = 4.2$:

1. Find the values of p and n.
2. Using the recurrence formula, determine the mode of X.
3. Compute the modal probability $P(X = m)$, where m is the mode, to 4 decimal places.

EXERCISE 14 **Uniform distribution from first principles**

A box contains tickets numbered 1 to 20. One ticket is drawn at random and X is the number on the ticket.

1. Show, from first principles, that $E(X) = 10.5$.
2. Show, from first principles, that $\mathrm{Var}(X) = 33.25$. (You may use $\sum_{k=1}^{n} k^2 = \dfrac{n(n+1)(2n+1)}{6}$.)
3. Find $P(X > 12)$ and $P(X \text{ is even})$.

EXERCISE 15 **Geometric distribution: the driving test**

The probability that a candidate passes a driving test at each attempt is 0.4, independently of previous attempts. Let X be the number of attempts needed to pass for the first time.

1. Find the probability that she passes at the third attempt.
2. Find the probability that she needs more than 4 attempts.

3. Find the expected number of attempts and the variance of X.
4. Find the smallest number n of attempts such that the probability of passing within n attempts exceeds 0.95.

EXERCISE 16 **Poisson: calls at an office and sum of independent variables**

Calls arrive at an office at random, at an average rate of 5 per hour.

1. Find the probability of exactly 3 calls in one hour.
2. Find the probability of no call in a 15-minute interval.
3. Find the probability of at least one call in a 10-minute interval.
4. Independently, emails arrive at an average rate of 3 per hour. Find the probability that the total number of calls and emails received in one hour is exactly 4, and the probability that this total is at least 2.

11.3 I prepare for the GCE

EXERCISE 17 **GCE-style: binomial tables and a critical value**

A biased coin has probability 0.3 of showing a head. It is tossed 8 times and X denotes the number of heads obtained.

1. State the distribution of X and write down $E(X)$ and $\mathrm{Var}(X)$. (3 marks)
2. Using cumulative binomial tables, find $P(X \geq 2)$. (2 marks)
3. Using cumulative binomial tables, find $P(1 \leq X \leq 4)$. (3 marks)
4. Using tables, or otherwise, find $P(X = 5)$. (2 marks)
5. Find the smallest integer r such that $P(X \leq r) > 0.9$. (2 marks)
6. Using the recurrence formula, determine the mode of X and compute the corresponding probability. (4 marks)

EXERCISE 18 **GCE-style: Poisson approximation to the binomial**

A factory in Douala produces 400 items per day, of which 1.5% are defective, indepen-

dently of one another. Let X be the number of defective items produced in a day.

1. State the exact distribution of X, and compute $E(X)$ and $\mathrm{Var}(X)$. (3 marks)
2. Explain why X may be approximated by a Poisson distribution, and state the parameter of this approximation. (2 marks)
3. Using this approximation, find the probability that exactly 6 defective items are produced in a day. (2 marks)
4. Using cumulative Poisson tables, find the probability that more than 8 defective items are produced in a day. (3 marks)
5. Find the probability that, in a day, the number of defective items lies between 3 and 9 inclusive. (3 marks)
6. On a randomly chosen day, the number of defective items exceeded 10. Comment on whether this casts doubt on the claimed defect rate of 1.5%. (2 marks)

EXERCISE 19 **GCE-style: from a frequency distribution to a Poisson model**

The table below gives the observed frequencies of the number of goals scored per match in 50 football matches of the national championship.

Number of goals x	0	1	2	3	4	5 or more
Observed frequency	8	15	13	9	4	1

1. Calculate the sample mean number of goals per match. (3 marks)
2. Explain why a Poisson distribution may be a suitable model, and estimate its parameter λ. (2 marks)
3. Using λ equal to your estimate (rounded to 1 decimal place), compute $P(X = 0)$, $P(X = 1)$ and $P(X = 2)$, and hence the expected frequencies for 0, 1 and 2 goals per match. (5 marks)
4. Compare the expected frequencies with the observed frequencies and comment on the goodness of fit of the Poisson model. (3 marks)
5. Using the model, find the probability that a randomly chosen match produces at least 3 goals. (2 marks)

12 SOLUTIONS TO THE EXERCISES

12.1 Solutions to "I check my knowledge"

SOLUTION **Exercise 1 — MCQ: spotting the right distribution**

1. Binomial: $X \sim B(20, 0.5)$. There is a fixed number $n = 20$ of independent trials, each with two outcomes (head/tail) and constant success probability $p = 0.5$.
2. Discrete uniform on $\{1, 2, \ldots, 50\}$: each ticket has the same probability $\frac{1}{50}$ of being drawn, so all outcomes are equally likely.
3. Geometric with parameter $p = 0.3$: we repeat independent identical trials until the first success (first goal), and X counts the number of trials needed.
4. Poisson with parameter $\lambda = 12$: events (arrivals) occur randomly, independently, at a constant average rate in a fixed interval of time (one hour).

SOLUTION **Exercise 2 — True or false?**

1. False. If $X \sim B(n, p)$ then $\mathrm{Var}(X) = np(1-p) = npq$, not np. It is $E(X)$ that equals np.
2. True. For $X \sim \mathrm{Po}(\lambda)$, $E(X) = \lambda$ and $\mathrm{Var}(X) = \lambda$; equality of mean and variance is the signature of the Poisson distribution.
3. True. $E(X) = \frac{1}{n}\sum_{k=1}^{n} k = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}$.
4. False. The sum of two independent Poisson variables is Poisson with parameter equal to the sum of the parameters: $X + Y \sim \mathrm{Po}(2+3) = \mathrm{Po}(5)$, not $\mathrm{Po}(6)$.

SOLUTION **Exercise 3 — Key definitions and formulas**

1. The four conditions of the Bernoulli scheme are:
   - a fixed number n of trials;
   - each trial has exactly two outcomes (success/failure);
   - the probability of success p is constant from trial to trial;
   - the trials are independent.

Then X = number of successes follows $B(n, p)$.

2. For $X \sim B(n,p)$: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, $k = 0, 1, \ldots, n$.

   For $Y \sim \text{Po}(\lambda)$: $P(Y = k) = e^{-\lambda}\dfrac{\lambda^k}{k!}$, $k = 0, 1, 2, \ldots$

3. The recurrence relation is
$$\frac{P(X = k+1)}{P(X = k)} = \frac{n-k}{k+1} \cdot \frac{p}{1-p}.$$
   The probabilities increase while $\dfrac{P(X = k+1)}{P(X = k)} > 1$, i.e. while $k < (n+1)p - 1$, and decrease afterwards. Hence the **mode** is $m = \lfloor (n+1)p \rfloor$ when $(n+1)p$ is not an integer; if $(n+1)p$ is an integer, there are two modes, $(n+1)p - 1$ and $(n+1)p$.

4. For the geometric distribution (number of trials up to and including the first success):
$$E(X) = \frac{1}{p}, \qquad \text{Var}(X) = \frac{1-p}{p^2} = \frac{q}{p^2}.$$

SOLUTION **Exercise 4 — Direct calculations**

1. Each throw is a Bernoulli trial with $p = P(\text{six}) = \frac{1}{6}$, repeated $n = 12$ times independently, so $X \sim B\left(12, \frac{1}{6}\right)$.
$$E(X) = np = 12 \times \tfrac{1}{6} = 2, \qquad \text{Var}(X) = npq = 12 \times \tfrac{1}{6} \times \tfrac{5}{6} = \frac{5}{3} \approx 1.6667.$$

2. $P(X = 2) = \binom{12}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^{10} = 66 \times \frac{1}{36} \times 0.16151 \approx \mathbf{0.2961}$.

3. $Y \sim \text{Po}(4)$:
$$P(Y = 0) = e^{-4} \approx \mathbf{0.0183}; \qquad P(Y = 3) = e^{-4}\frac{4^3}{3!} = e^{-4} \times \frac{64}{6} \approx \mathbf{0.1954}.$$

4. T uniform on $\{1, \ldots, 10\}$:
$$E(T) = \frac{n+1}{2} = \frac{11}{2} = 5.5, \qquad \text{Var}(T) = \frac{n^2-1}{12} = \frac{100-1}{12} = \frac{99}{12} = 8.25.$$

12.2 Solutions to "I practise"

SOLUTION **Exercise 5 — Binomial: parameters and mode from the moments**

1. We have $np = 6$ and $npq = 4.2$. Dividing: $q = \frac{npq}{np} = \frac{4.2}{6} = 0.7$, so $p = 1 - q = 0.3$ and $n = \frac{6}{0.3} = 20$. Hence $X \sim B(20, 0.3)$.
2. Compute $(n+1)p = 21 \times 0.3 = 6.3$, not an integer. By the recurrence relation, the probabilities increase while $k < (n+1)p - 1 = 5.3$ and decrease from $k = 6$ onwards. Hence the mode is $m = \lfloor 6.3 \rfloor = 6$.
3. $P(X = 6) = \binom{20}{6}(0.3)^6(0.7)^{14} = 38760 \times 0.000729 \times 0.0067822 \approx \mathbf{0.1916}$.

SOLUTION **Exercise 6 — Uniform distribution from first principles**

Each value $k \in \{1, \dots, 20\}$ has probability $\frac{1}{20}$.

1. Using $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ with $n = 20$:

$$E(X) = \sum_{k=1}^{20} k \cdot \frac{1}{20} = \frac{1}{20} \cdot \frac{20 \times 21}{2} = \frac{210}{20} = 10.5. \quad \blacksquare$$

2. Using $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ with $n = 20$:

$$E(X^2) = \sum_{k=1}^{20} k^2 \cdot \frac{1}{20} = \frac{1}{20} \times \frac{20 \times 21 \times 41}{6} = \frac{21 \times 41}{6} = \frac{861}{6} = 143.5.$$

Then $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 143.5 - 10.5^2 = 143.5 - 110.25 = 33.25$. $\blacksquare$

Check with the standard formula: $\mathrm{Var}(X) = \frac{n^2 - 1}{12} = \frac{400 - 1}{12} = \frac{399}{12} = 33.25$. ✓

3. $P(X > 12) = P(X \in \{13, \dots, 20\}) = \frac{8}{20} = 0.4$.

The even values are $\{2, 4, \dots, 20\}$, i.e. 10 values: $P(X \text{ even}) = \frac{10}{20} = 0.5$.

SOLUTION **Exercise 7 — Geometric distribution: the driving test**

$X \sim$ geometric with $p = 0.4$, $q = 0.6$: $P(X = k) = q^{k-1}p$.

1. $P(X = 3) = (0.6)^2 \times 0.4 = 0.36 \times 0.4 = \mathbf{0.144}$.
2. "More than 4 attempts" means the first 4 attempts all fail:
$$P(X > 4) = q^4 = (0.6)^4 = \mathbf{0.1296}.$$
3. $E(X) = \frac{1}{p} = \frac{1}{0.4} = 2.5$ attempts; $\quad \mathrm{Var}(X) = \frac{q}{p^2} = \frac{0.6}{0.16} = 3.75$.
4. $P(X \le n) = 1 - q^n = 1 - (0.6)^n > 0.95 \iff (0.6)^n < 0.05$. Taking logarithms: $n \ln 0.6 < \ln 0.05$, and since $\ln 0.6 < 0$ the inequality reverses on division:
$$n > \frac{\ln 0.05}{\ln 0.6} \approx \frac{-2.9957}{-0.5108} \approx 5.86.$$
The smallest integer is $\mathbf{n = 6}$. Check: $(0.6)^6 = 0.0467 < 0.05$, so $P(X \le 6) = 0.9533 > 0.95$. ✓

SOLUTION **Exercise 8 — Poisson: calls at an office and sum of independent variables**

Calls: rate 5 per hour, so in one hour $X \sim \mathrm{Po}(5)$.

1. $P(X = 3) = e^{-5}\frac{5^3}{3!} = e^{-5} \times \frac{125}{6} \approx \mathbf{0.1404}$.
2. In 15 minutes the mean is $\lambda = 5 \times \frac{15}{60} = 1.25$ (the parameter is proportional to the length of the interval — the unit interval must be adjusted). With $N \sim \mathrm{Po}(1.25)$:
$$P(N = 0) = e^{-1.25} \approx \mathbf{0.2865}.$$
3. In 10 minutes, $\lambda = 5 \times \frac{10}{60} = \frac{5}{6} \approx 0.8333$. With $M \sim \mathrm{Po}(0.8333)$:
$$P(M \ge 1) = 1 - P(M = 0) = 1 - e^{-0.8333} \approx 1 - 0.4346 = \mathbf{0.5654}.$$
4. Emails: $Y \sim \mathrm{Po}(3)$ per hour, independent of $X \sim \mathrm{Po}(5)$. By the additive property of independent Poisson variables, $X + Y \sim \mathrm{Po}(5 + 3) = \mathrm{Po}(8)$.
$$P(X + Y = 4) = e^{-8}\frac{8^4}{4!} = e^{-8} \times \frac{4096}{24} \approx \mathbf{0.0573}.$$
$$P(X + Y \ge 2) = 1 - P(0) - P(1) = 1 - e^{-8}(1 + 8) = 1 - 9e^{-8} \approx 1 - 0.0030 = \mathbf{0.9970}.$$

12.3 Solutions to "I prepare for the GCE"

SOLUTION **Exercise 9 — GCE-style: binomial tables and a critical value**

1. $X \sim B(8, 0.3)$ **(B1)**; $E(X) = np = 8 \times 0.3 = 2.4$ **(M1)**; $\mathrm{Var}(X) = npq = 8 \times 0.3 \times 0.7 = 1.68$ **(A1)**.

2. From cumulative tables for $B(8, 0.3)$: $P(X \le 1) = 0.2553$.

$$P(X \ge 2) = 1 - P(X \le 1) = 1 - 0.2553 = \mathbf{0.7447} \quad \text{(M1 A1)}.$$

3. $P(1 \le X \le 4) = P(X \le 4) - P(X \le 0) = 0.9420 - 0.0576 = \mathbf{0.8844}$ (M1 for the difference, A1 for each value used).

4. $P(X = 5) = P(X \le 5) - P(X \le 4) = 0.9887 - 0.9420 = \mathbf{0.0467}$ (M1 A1).
   Check by formula: $\binom{8}{5}(0.3)^5(0.7)^3 = 56 \times 0.00243 \times 0.343 = 0.0467$. ✓

5. From the tables: $P(X \le 3) = 0.8059 \le 0.9$ and $P(X \le 4) = 0.9420 > 0.9$. Hence the smallest integer is **r = 4** (M1 A1).

6. $(n + 1)p = 9 \times 0.3 = 2.7$, not an integer, so the mode is $m = \lfloor 2.7 \rfloor = 2$ (M1 A1). Equivalently, the ratio $\dfrac{P(X = k + 1)}{P(X = k)} = \dfrac{8 - k}{k + 1} \cdot \dfrac{3}{7}$ is > 1 for $k = 0, 1$ and < 1 for $k \ge 2$.

$$P(X = 2) = \binom{8}{2}(0.3)^2(0.7)^6 = 28 \times 0.09 \times 0.117649 \approx \mathbf{0.2965} \quad \text{(M1 A1)}.$$

Examiner's expectations: correct reading of cumulative tables (subtracting the right cumulative value), clear statement of the distribution with both parameters, and answers given to 4 decimal places. A common error is writing $P(X \ge 2) = 1 - P(X \le 2)$: this loses the method mark.

SOLUTION **Exercise 10 — GCE-style: Poisson approximation to the binomial**

1. $X \sim B(400, 0.015)$ **(B1)**; $E(X) = np = 400 \times 0.015 = 6$ **(A1)**; $\mathrm{Var}(X) = npq = 6 \times 0.985 = 5.91$ **(A1)**.

2. $n = 400$ is large and $p = 0.015$ is small (with $np = 6$ moderate), so the Poisson approximation is valid (M1): $X \approx \mathrm{Po}(\lambda)$ with $\lambda = np = 6$ (A1).

3. $P(X = 6) \approx e^{-6}\dfrac{6^6}{6!} = e^{-6} \times \dfrac{46656}{720} \approx \mathbf{0.1606}$ (M1 A1).

4. From cumulative Poisson tables with $\lambda = 6$: $P(X \le 8) = 0.8472$.

$$P(X > 8) = 1 - P(X \le 8) = 1 - 0.8472 = \mathbf{0.1528} \quad \text{(M1 A1 A1)}.$$

5. $P(3 \le X \le 9) = P(X \le 9) - P(X \le 2) = 0.9161 - 0.0620 = \mathbf{0.8541}$ (M1 A1 A1).

6. $P(X > 10) = 1 - P(X \leq 10) = 1 - 0.9574 = 0.0426$ (M1). Observing more than 10 defectives has probability about 4.3% under the model: this is a rare event (less than 5%), so it raises some doubt about the claimed rate of 1.5% — the true rate may be higher — although such an event can still occur by chance; further days should be monitored before concluding (A1).

Examiner's expectations: the conditions for the approximation must be stated (n large, p small, np moderate); in (f) a numerical probability must support the comment — a comment without computation earns no mark.

SOLUTION **Exercise 11 — GCE-style: from a frequency distribution to a Poisson model**

1. Total goals: $0(8) + 1(15) + 2(13) + 3(9) + 4(4) + 5(1) = 15 + 26 + 27 + 16 + 5 = 89$ (M1).

$$\bar{x} = \frac{89}{50} = 1.78 \text{ goals per match} \quad \text{(A1 A1)}.$$

2. Goals are rare, random events occurring independently at an approximately constant average rate per match, so a Poisson model is appropriate (B1). We estimate λ by the sample mean (since $E(X) = \lambda$ for a Poisson distribution): $\hat{\lambda} = 1.78 \approx 1.8$ (B1).

3. With $\lambda = 1.8$:

$$\begin{aligned} P(X = 0) &= e^{-1.8} \approx 0.1653 && \Rightarrow \quad 50 \times 0.1653 \approx 8.3, \\ P(X = 1) &= 1.8\, e^{-1.8} \approx 0.2975 && \Rightarrow \quad 50 \times 0.2975 \approx 14.9, \\ P(X = 2) &= e^{-1.8} \frac{1.8^2}{2} \approx 0.2678 && \Rightarrow \quad 50 \times 0.2678 \approx 13.4. \end{aligned}$$

(M1 for the pmf, A1 for each probability, A1 for the expected frequencies.)

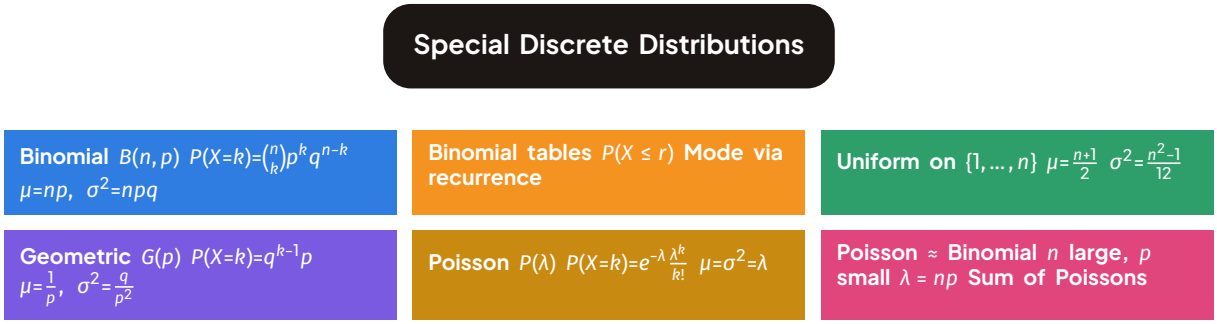
Goals x	0	1	2
Observed frequency	8	15	13
Expected frequency	8.3	14.9	13.4

4. The expected frequencies are very close to the observed ones (differences of at most 0.4) (M1 A1): the Poisson model with $\lambda = 1.8$ fits the data very well (A1).

5. $P(X \geq 3) = 1 - [P(0) + P(1) + P(2)] = 1 - (0.1653 + 0.2975 + 0.2678) = 1 - 0.7306 = \mathbf{0.2694}$ (M1 A1).

Examiner's expectations: the mean must be computed with full working from the frequency table; the estimate $\hat{\lambda} = \bar{x}$ must be justified (the mean of a Poisson distribution is λ); expected frequencies are obtained by multiplying each probability by the total frequency 50; the comparison must be quantitative, not just "the fit is good".

13 SUMMARY SHEET



Mind map of the chapter: the special discrete probability distributions.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- **Binomial distribution** $X \sim B(n,p)$: number of successes in n independent identical trials, each with $P(\text{success}) = p$, $q = 1-p$.
- **Discrete uniform distribution** on $\{1, 2, \dots, n\}$: every outcome equally likely, $P(X = k) = \frac{1}{n}$.
- **Geometric distribution** $X \sim G(p)$: number of trials up to and including the first success; memoryless.
- **Poisson distribution** $X \sim P(\lambda)$: number of rare events in a fixed interval of time/space, rate λ per unit interval.
- **Mode**: the value of k with the greatest probability.

Formulas to know by heart

- Binomial: $P(X = k) = \binom{n}{k} p^k q^{n-k}$, $E(X) = np$, $\operatorname{Var}(X) = npq$.
- Recurrence: $\frac{P(X = k+1)}{P(X = k)} = \frac{n-k}{k+1} \cdot \frac{p}{q}$; the mode is the integer part of $(n+1)p$ (if $(n+1)p$ is an integer, both $(n+1)p$ and $(n+1)p - 1$ are modes).
- Uniform: $E(X) = \frac{n+1}{2}$, $\operatorname{Var}(X) = \frac{n^2-1}{12}$.
- Geometric: $P(X = k) = q^{k-1}p$, $P(X > k) = q^k$, $E(X) = \frac{1}{p}$, $\operatorname{Var}(X) = \frac{q}{p^2}$.
- Poisson: $P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$, $E(X) = \operatorname{Var}(X) = \lambda$.
- Scaling: if λ is the rate per unit interval, the rate over an interval of length t is λt.

- Sum of independent Poissons: $X \sim P(\lambda_1)$, $Y \sim P(\lambda_2)$ independent $\Rightarrow X + Y \sim P(\lambda_1 + \lambda_2)$.
- Poisson approximation: $B(n, p) \approx P(np)$ when n is large ($n \geq 50$) and p small ($p \leq 0.1$), so that $\lambda = np$ is moderate.

Using the tables

- Cumulative binomial tables give $P(X \leq r)$ directly. Then: $P(X \geq r) = 1 - P(X \leq r - 1)$; $P(X = r) = P(X \leq r) - P(X \leq r - 1)$; $P(a \leq X \leq b) = P(X \leq b) - P(X \leq a - 1)$.
- Cumulative Poisson tables give $P(X \leq r)$ for tabulated λ; use the same transformations.
- If $p > 0.5$ for the binomial, work with the number of failures $Y \sim B(n, 1 - p)$.

Methods

- **Identify the model first**: fixed number of trials → binomial; waiting for first success → geometric; random rare events in an interval → Poisson; equally likely scores → uniform.
- **Mode of a binomial**: compute $\frac{P(X = k+1)}{P(X = k)} \geq 1 \iff k \leq (n+1)p - 1$; probabilities increase up to the mode then decrease.
- **Frequency link**: expected frequency = (total frequency) $\times P(X = k)$; compare with observed frequencies to judge goodness of fit.

Common mistakes

- Confusing $P(X \leq r)$ with $P(X < r)$: $P(X < r) = P(X \leq r - 1)$.
- Forgetting $q = 1 - p$, or writing q^{n-k} with the wrong exponent.
- Using the Poisson approximation when p is not small.
- For the geometric law, forgetting that X counts the successful trial too: $P(X = k) = q^{k-1}p$, not $q^k p$.
- Forgetting to adjust λ when the interval changes (e.g. per hour → per 15 minutes: divide by 4).
- Adding Poisson variables that are not independent.

GCE tips

- Quote the distribution and parameters first: "$X \sim B(10, 0.3)$" earns the first mark.
- Show the formula before substituting; examiners reward method marks.
- Give final probabilities to 4 decimal places (as in the tables).
- For "mean and variance" questions, state $E(X)$ and $\mathrm{Var}(X)$, not the standard deviation, unless asked.
- Always check: probabilities must lie in $[0, 1]$ and $\sum P(X = k) = 1$.

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